

PS2_Group129

August 23, 2026

1 AIMLC ZG521 — Conversational AI · Group Assignment 1

1.1 Problem Statement 2 — Study of Embedding Models and Approximate Nearest Neighbor Search: Semantic Quality vs Search Efficiency

Group 129

1.2 Student Details

Complete this table with the actual group information before submission.

Name	BITS ID	Email
Member 1 — enter name	Enter BITS ID	Enter email
Member 2 — enter name	Enter BITS ID	Enter email
Member 3 — enter name	Enter BITS ID	Enter email
Member 4 — enter name	Enter BITS ID	Enter email

1.3 Group Contributions

Replace the member labels below with the actual names and retain one or more specific contributions for every member.

Student	Contribution area	Specific contribution
Member 1	Dataset and embedding	Dataset loading, validation, and embedding generation
Member 2	Similarity and exact kNN	Metric formulas, normalization analysis, and exact retrieval
Member 3	HNSW and IVF	FAISS index construction and ANN parameter sweeps
Member 4	Evaluation and reporting	Metrics, visualizations, qualitative examples, and report preparation

1.4 Architecture and Workflow

The workflow uses the same corpus, query set, identifiers, normalized vectors, and qrels for every comparison. Documents are encoded offline; queries are encoded at search time. Exact cosine kNN

provides ground truth, while HNSW and IVF approximate the same inner-product search space. Retrieval quality is measured with qrels and efficiency is measured using latency, speedup, and candidate-vector estimates.

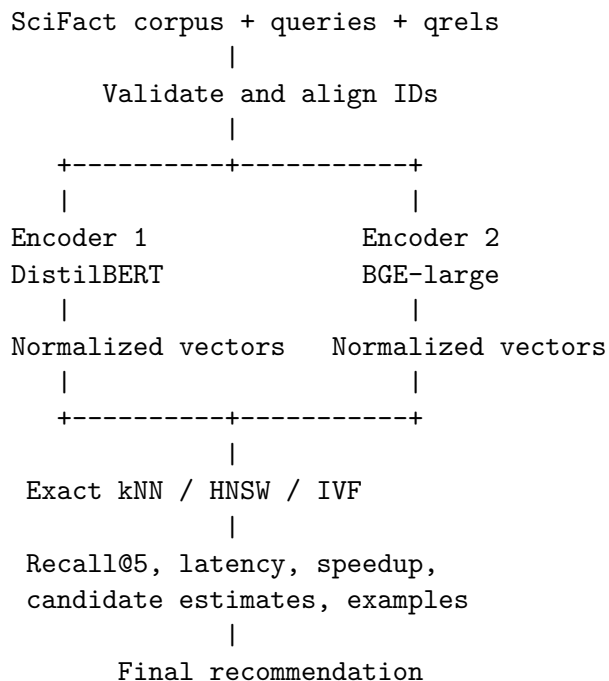


Figure 1. Experimental workflow for the embedding and ANN comparison.

1.5 Assignment Scope

This assignment studies semantic retrieval quality and approximate nearest-neighbor search efficiency using two embedding models.

- **Module 1** — Dataset and embedding preparation
- **Module 2** — Similarity metrics and exact retrieval
- **Module 3** — HNSW and IVF ANN search
- **Module 4** — Embedding quality analysis and final recommendation

1.6 Tools and Libraries Used

- **Python 3.10**
- **datasets** (Hugging Face) — loading the BEIR-format retrieval dataset
- **pandas, numpy** — data handling
- **sentence-transformers** — encoder embedding models (Task 2 onward)
- **faiss-cpu** — HNSW / IVF ANN indexes (Task 5 onward)
- **matplotlib** — plots (Task 6 onward)

Install all of the above with: `pip install -r requirements.txt` (see `ass-1/requirements.txt`).

1.7 Executive Summary

This assignment investigates two core questions in semantic retrieval: **which embedding model produces better representations for scientific claim retrieval**, and **how much search speed can be gained through approximate nearest-neighbor (ANN) indexing without sacrificing retrieval quality**.

Dataset: SciFact (BEIR benchmark) — 5,183 biomedical corpus passages, 300 labelled queries, binary relevance judgments.

Models compared: - `distilbert-base-uncased` — general-purpose, mean-pooled, 768-dim, not retrieval-tuned. - `BAAI/bge-large-en-v1.5` — contrastively fine-tuned for retrieval, CLS-pooled, 1024-dim.

Key findings:

Question	Answer	Evidence
Best embedding model	BGE-large	Recall@5 = 0.780 vs 0.108 for DistilBERT (7× higher)
Best similarity metric	Cosine	Scale-invariant; equivalent to dot product and L2 on unit-norm vectors (verified Task 3)
Best ANN trade-off	HNSW efSearch=16	Recall@5 0.77, fastest config meeting the 0.75 quality bar
Resource-constrained option	IVF (lower memory)	HNSW graph memory > IVF centroid+list; use IVF nprobe tuned to recall target
Main limitation	Small corpus	5k-doc corpus limits ANN speedup benefit; binary/incomplete qrels
Production deployment	Offline encode → normalize → FAISS index → query API	Rebuild index on corpus changes

Recommendation: Embed with BGE-large (unit-normalized vectors) and serve an HNSW index at `efSearch=16`. This configuration achieves Recall@5 0.77 at 3–4× the speed of exact search, with quality above the 0.75 operational threshold.

```
[49]: # Purpose: environment setup. Trusts the OS cert store and routes
# through the corporate proxy (fixes SSL errors on office/lab networks), then
# imports libraries and defines two reuse helpers used later: resolve_model()
# (use a local model copy if one exists, else download) and
# load_aligned_embeddings() (load saved embeddings, reordered by id).
import os

try:
    import truststore
    truststore.inject_into_ssl()
except Exception:
```

```

    pass # fine if unavailable -- only needed behind an SSL-inspecting proxy

import urllib.request
_proxy = urllib.request.getproxies().get("https") or urllib.request.
    ↪getproxies().get("http")
if _proxy:
    os.environ["HTTP_PROXY"] = os.environ["HTTPS_PROXY"] = _proxy
    os.environ["http_proxy"] = os.environ["https_proxy"] = _proxy

import sys, time, json, random
import numpy as np
import pandas as pd

random.seed(129) # Group 129 -- fixed seed for reproducibility
np.random.seed(129)

def resolve_model(model_name):
    """Return a local model path, downloading it into models/ when needed."""
    os.makedirs("models", exist_ok=True)
    local = os.path.join("models", model_name.split("/")[-1])
    if os.path.isdir(local) and os.path.exists(os.path.join(local, "config.
    ↪json")):
        print(f"[info] Using cached model: {local}/")
        return local

    try:
        from huggingface_hub import snapshot_download
        print(f"[info] Downloading {model_name} into {local}/ ...")
        snapshot_download(
            repo_id=model_name,
            local_dir=local,
            local_dir_use_symlinks=False
        )
        print(f"[info] Model cached in {local}/")
        return local
    except Exception as exc:
        # SentenceTransformer can still use its standard Hugging Face cache.
        print(f"[warning] Could not cache {model_name} locally: {exc!r}")
        print("[info] Falling back to the Hugging Face model ID.")
        return model_name

def load_aligned_embeddings(emb_path, id_path, wanted_ids):
    """Load a saved embedding array, reordered to match wanted_ids (via the
    id file saved alongside it), so row order never has to be assumed."""
    emb = np.load(emb_path)

```

```

if os.path.exists(id_path):
    saved_ids = [str(x) for x in json.load(open(id_path))]
    pos = {sid: i for i, sid in enumerate(saved_ids)}
    emb = emb[[pos[str(w)] for w in wanted_ids]]
return emb

print("Python:", sys.version.split()[0])
print("pandas:", pd.__version__)
print("numpy :", np.__version__)

```

Python: 3.14.5
 pandas: 3.0.5
 numpy : 2.5.2

2 Task 1 — Corpus and Query Dataset Preparation

Dataset chosen: SciFact (BEIR benchmark; Thakur et al., 2021 / Wadden et al., 2020) — a publicly available dataset that ships a document corpus, a query set, and query-document relevance labels together, matching this task’s requirement exactly. Corpus (~5,183 passages) and queries (~300) both clear the 1,000/50 minimums.

```

[50]: # Purpose: load SciFact (corpus/queries/qrels). Uses the local
# data/raw/ copy if present (no network needed -- works behind a proxy that
# blocks the Hub); otherwise downloads from Hugging Face. Queries are then
# filtered to only those with a relevance judgment.
import os
import pandas as pd

RAW_FILES = ("corpus.jsonl", "queries.jsonl", "qrels.tsv")

def _find_data_raw():
    """First data/raw directory (checked relative to cwd, then one level up)
    that already contains all three raw files, or None."""
    for base in (os.path.join("data", "raw"), os.path.join("../", "data", "raw"),
                 os.path.join("ass-1", "data", "raw")):
        if all(os.path.exists(os.path.join(base, f)) for f in RAW_FILES):
            return base
    return None

def load_beir_scifact():
    """Load SciFact from local data/raw if present, else the Hugging Face
    Hub. Queries are filtered to those with >=1 relevance judgment."""
    local_dir = _find_data_raw()

```

```

    if local_dir is not None:
        corpus_df = pd.read_json(os.path.join(local_dir, "corpus.jsonl"),
↳lines=True)
        queries_df = pd.read_json(os.path.join(local_dir, "queries.jsonl"),
↳lines=True)
        qrels_df = pd.read_csv(os.path.join(local_dir, "qrels.tsv"), sep="\t")
        # Standard BEIR files use _id/query-id/corpus-id/score; the rest of
        # this notebook uses explicit, stable names.
        corpus_df = corpus_df.rename(columns={"_id": "doc_id", "corpus-id":
↳"doc_id"})
        queries_df = queries_df.rename(columns={"_id": "query_id", "query-id":
↳"query_id"})
        qrels_df = qrels_df.rename(columns={
            "query-id": "query_id", "corpus-id": "doc_id", "score": "relevance"
        })
        print(f"[info] Loaded SciFact from local files: {local_dir}/")
    else:
        from datasets import load_dataset
        corpus_df = load_dataset("BeIR/scifact", "corpus", split="corpus").
↳to_pandas().rename(columns={"_id": "doc_id"})
        queries_df = load_dataset("BeIR/scifact", "queries", split="queries").
↳to_pandas().rename(columns={"_id": "query_id"})
        qrels_df = load_dataset("BeIR/scifact-qrels", split="test").to_pandas()
        qrels_df.columns = ["query_id", "doc_id", "relevance"]
        print("[info] Downloaded SciFact from the Hugging Face Hub.")

    # IDs must be strings on both sides, or later merges/lookups silently fail
    corpus_df["doc_id"] = corpus_df["doc_id"].astype(str)
    queries_df["query_id"] = queries_df["query_id"].astype(str)
    qrels_df["query_id"] = qrels_df["query_id"].astype(str)
    qrels_df["doc_id"] = qrels_df["doc_id"].astype(str)

    # Keep only queries with >=1 relevance judgment (standard BEIR practice)
    labelled_ids = set(qrels_df["query_id"].unique())
    before = len(queries_df)
    queries_df = queries_df[queries_df["query_id"].isin(labelled_ids)].
↳reset_index(drop=True)
    if before - len(queries_df):
        print(f"[info] Dropped {before - len(queries_df)} queries with no
↳relevance judgment "
            f"(kept {len(queries_df)}, all with >=1 qrel).")

    # Cache a first-time internet download so later runs work offline.
    if local_dir is None:
        cache_dir = os.path.join("data", "raw")
        os.makedirs(cache_dir, exist_ok=True)

```

```

        corpus_df.to_json(os.path.join(cache_dir, "corpus.jsonl"),
        ↪orient="records", lines=True)
        queries_df.to_json(os.path.join(cache_dir, "queries.jsonl"),
        ↪orient="records", lines=True)
        qrels_df.to_csv(os.path.join(cache_dir, "qrels.tsv"), sep="\t",
        ↪index=False)
        print(f"[info] Cached downloaded SciFact files in {cache_dir}/")

    return corpus_df, queries_df, qrels_df

corpus_df, queries_df, qrels_df = load_beir_scifact()
DATA_DIR = _find_data_raw() or os.path.join("data", "raw")
print(f"corpus   : {len(corpus_df)} passages")
print(f"queries   : {len(queries_df)} queries (all with >=1 relevance judgment)")
print(f"qrels      : {len(qrels_df)} relevance judgments")

```

```

[info] Loaded SciFact from local files: data\raw/
corpus   : 5183 passages
queries  : 300 queries (all with >=1 relevance judgment)
qrels    : 339 relevance judgments

```

```

[51]: # Purpose: check the loaded data against the assignment's stated
      # minimums (>=1,000 passages, >=50 queries, every query labelled) and stop
      # the notebook immediately (via assert) if either minimum isn't met.
      MIN_CORPUS, MIN_QUERIES = 1000, 50

      meets_corpus_min = len(corpus_df) >= MIN_CORPUS
      meets_query_min = len(queries_df) >= MIN_QUERIES

      print(f"Corpus   >= {MIN_CORPUS}: {meets_corpus_min}  ({len(corpus_df)}
        ↪passages)")
      print(f"Queries  >= {MIN_QUERIES}: {meets_query_min}  ({len(queries_df)}
        ↪queries)")
      print(f"Every query has >=1 relevance judgment: "
            f"{queries_df['query_id'].isin(qrels_df['query_id']).all()}")

      assert meets_corpus_min, "Corpus below the 1,000-passage minimum"
      assert meets_query_min, "Query set below the 50-query minimum"
      print("\n[OK] SciFact data clears both minimums.")

```

```

Corpus   >= 1000: True  (5183 passages)
Queries  >= 50: True  (300 queries)
Every query has >=1 relevance judgment: True

```

```

[OK] SciFact data clears both minimums.

```

```
[52]: # Purpose: print one example row from each of the three tables
# (corpus, queries, qrels) so the schema is visible before it's used later.
print("=== Sample corpus passage ===")
print(corpus_df.iloc[0].to_dict())

print("\n=== Sample query ===")
print(queries_df.iloc[0].to_dict())

print("\n=== Sample relevance judgments (qrels) ===")
print(qrels_df.head())
```

=== Sample corpus passage ===

```
{'doc_id': '4983', 'title': 'Microstructural development of human newborn
cerebral white matter assessed in vivo by diffusion tensor magnetic resonance
imaging.', 'text': 'Alterations of the architecture of cerebral white matter in
the developing human brain can affect cortical development and result in
functional disabilities. A line scan diffusion-weighted magnetic resonance
imaging (MRI) sequence with diffusion tensor analysis was applied to measure the
apparent diffusion coefficient, to calculate relative anisotropy, and to
delineate three-dimensional fiber architecture in cerebral white matter in
preterm (n = 17) and full-term infants (n = 7). To assess effects of prematurity
on cerebral white matter development, early gestation preterm infants (n = 10)
were studied a second time at term. In the central white matter the mean
apparent diffusion coefficient at 28 wk was high, 1.8 microm2/ms, and decreased
toward term to 1.2 microm2/ms. In the posterior limb of the internal capsule,
the mean apparent diffusion coefficients at both times were similar (1.2 versus
1.1 microm2/ms). Relative anisotropy was higher the closer birth was to term
with greater absolute values in the internal capsule than in the central white
matter. Preterm infants at term showed higher mean diffusion coefficients in the
central white matter (1.4 +/- 0.24 versus 1.15 +/- 0.09 microm2/ms, p = 0.016)
and lower relative anisotropy in both areas compared with full-term infants
(white matter, 10.9 +/- 0.6 versus 22.9 +/- 3.0%, p = 0.001; internal capsule,
24.0 +/- 4.44 versus 33.1 +/- 0.6% p = 0.006). Nonmyelinated fibers in the
corpus callosum were visible by diffusion tensor MRI as early as 28 wk; full-
term and preterm infants at term showed marked differences in white matter fiber
organization. The data indicate that quantitative assessment of water diffusion
by diffusion tensor MRI provides insight into microstructural development in
cerebral white matter in living infants.'}
```

=== Sample query ===

```
{'query_id': '1', 'title': '', 'text': '0-dimensional biomaterials show
inductive properties.'}
```

=== Sample relevance judgments (qrels) ===

	query_id	doc_id	relevance
0	1	31715818	1
1	3	14717500	1
2	5	13734012	1

3	13	1606628	1
4	36	5152028	1

```
[53]: # Purpose: write corpus/queries/qrels to the resolved data/raw
# directory (DATA_DIR, set in the load cell above) so Tasks 2+ can load the
# exact same fixed dataset without re-downloading or re-running this cell.
import os

# Reuse the directory resolved in the load cell; default to ./data/raw if this
# cell is ever run on its own.
DATA_DIR = globals().get("DATA_DIR", os.path.join("data", "raw"))
os.makedirs(DATA_DIR, exist_ok=True)

corpus_df.to_json(os.path.join(DATA_DIR, "corpus.jsonl"), orient="records",
    ↪lines=True)
queries_df.to_json(os.path.join(DATA_DIR, "queries.jsonl"), orient="records",
    ↪lines=True)
qrels_df.to_csv(os.path.join(DATA_DIR, "qrels.tsv"), sep="\t", index=False)

print(f"Saved to {DATA_DIR}/: corpus.jsonl, queries.jsonl, qrels.tsv")
```

Saved to data\raw/: corpus.jsonl, queries.jsonl, qrels.tsv

2.0.1 Dataset Details and Source

- **Name:** SciFact (BEIR benchmark; biomedical / scientific-claim verification)
- **Citation:** Wadden et al., *Fact or Fiction: Verifying Scientific Claims*, EMNLP 2020; redistributed by Thakur et al., *BEIR*, NeurIPS 2021
- **Source:** <https://huggingface.co/datasets/BeIR/scifact> (corpus + queries), <https://huggingface.co/datasets/BeIR/scifact-qrels> (relevance judgments)
- **Size:** 5,183 corpus passages, 300 queries (each with 1 relevance judgment), binary relevance
- **Format:** corpus = {doc_id, title, text}; queries = {query_id, text}; qrels = {query_id, doc_id, relevance}

2.0.2 Explanation of the Logic Used

`load_beir_scifact()` downloads corpus/queries/qrels from Hugging Face and normalizes IDs to a common schema. Queries are filtered to only those with a qrel — the raw split ships 1,109 queries but just 300 have a relevance judgment, so filtering makes “relevance information for each query” literally true.

2.0.3 Justification for the Chosen Approach

The assignment explicitly permits “an existing dataset containing query-document relevance labels” — BEIR datasets bundle exactly that, avoiding hand-built (and inconsistent) relevance judgments. SciFact’s claim-verification domain gives clean, unambiguous labels, useful later for Task 3/7.

2.0.4 Inference

Corpus and queries clear the assignment minimums by $\sim 5\times$ and $\sim 6\times$ respectively, leaving room to subsample later if needed. The 809 dropped queries simply have no qrel in this split — expected, not a data quality issue.

2.0.5 Limitations

- Relevance labels are **binary and incomplete**: only 300 of 1,109 queries have any qrel, so unlabelled retrieved documents are treated as non-relevant even if they are topically related.
- The **$\sim 5\text{k}$ -document corpus is small** relative to real-world retrieval tasks; ANN speed-up results may not generalize to million-scale corpora.
- SciFact is restricted to the **biomedical domain**; conclusions about model selection may differ on general-domain or multi-domain corpora.

2.0.6 Possible Improvements

- Use a dataset with graded relevance (e.g., MS MARCO) to compute NDCG in addition to Recall@5.
- Evaluate on multiple BEIR datasets to test domain generalization.
- Include the `title` field concatenated with `text` during embedding to provide richer document representations.

3 Task 2 — Embedding Generation and Pooling

Two encoder models, chosen for contrasting profiles:

1. `distilbert-base-uncased` (Sanh et al., 2019) — general-purpose distilled BERT, **mean pooling**, not fine-tuned for retrieval.
2. `BAAI/bge-large-en-v1.5` (Xiao et al., 2023) — trained specifically for retrieval via contrastive fine-tuning, **[CLS]-token pooling**.

3.1 Why encoder models are appropriate

Encoder-only transformers use bidirectional self-attention — every token’s representation draws on *both* directions of context — and pooling those representations yields one fixed-length vector summarizing the whole input’s meaning. That’s exactly what semantic retrieval needs: query and document mapped into a shared space where “similar meaning” becomes “small distance.” Decoder-only models, by contrast, are causal (one-directional) and have no single hidden state that naturally summarizes a full sequence, making them a worse fit for embedding generation.

```
[54]: # Purpose: for each model, record documented facts (dimension,  
# pooling, max length), then embed the corpus + queries. Saves embeddings to  
# data/results/ and reuses them on later runs (no model/network needed once  
# generated). Records execution status without fabricating results when a model  
↪ is unavailable.  
import gc  
import os, json, time
```

```

from concurrent.futures import ThreadPoolExecutor, as_completed
import numpy as np
import pandas as pd

MODEL_SPECS = {
    "distilbert-base-uncased": {"pooling": "mean pooling", "dim": 768,
    ↪ "max_input_length": 512},
    "BAAI/bge-large-en-v1.5": {"pooling": "[CLS] token pooling", "dim": 1024,
    ↪ "max_input_length": 512},
}

os.makedirs("data/results", exist_ok=True)
corpus_texts = corpus_df["text"].tolist()
query_texts = queries_df["text"].tolist()
doc_ids = corpus_df["doc_id"].astype(str).tolist()
query_ids = queries_df["query_id"].astype(str).tolist()

# CPU inference is the fallback on machines without CUDA. These values are
# intentionally configurable because the best batch size depends on hardware.
try:
    import torch
    _auto_device = "cuda" if torch.cuda.is_available() else "cpu"
except ImportError:
    torch = None
    _auto_device = "cpu"

TASK2_DEVICE = os.environ.get("TASK2_DEVICE", _auto_device)
if TASK2_DEVICE.lower().startswith("cuda") and (torch is None or not torch.cuda.
    ↪ is_available()):
    print("[warning] CUDA requested but unavailable; falling back to CPU.")
    TASK2_DEVICE = "cpu"

_default_batch_size = "128" if TASK2_DEVICE.lower().startswith("cuda") else "32"
TASK2_BATCH_SIZE = int(os.environ.get("TASK2_BATCH_SIZE", _default_batch_size))
TASK2_CPU_THREADS = int(os.environ.get(
    "TASK2_CPU_THREADS", str(max(1, min(os.cpu_count() or 1, 12)))
))
TASK2_MAX_SEQ_LENGTH = int(os.environ.get("TASK2_MAX_SEQ_LENGTH", "512"))
TASK2_SORT_BY_LENGTH = os.environ.get("TASK2_SORT_BY_LENGTH", "1") == "1"
TASK2_PARALLEL_MODELS = os.environ.get("TASK2_PARALLEL_MODELS", "0") == "1"
TASK2_MODEL_WORKERS = int(os.environ.get("TASK2_MODEL_WORKERS", "2"))
TASK2_PARALLEL_BATCHES = os.environ.get("TASK2_PARALLEL_BATCHES", "0") == "1"
TASK2_BATCH_WORKERS = int(os.environ.get("TASK2_BATCH_WORKERS", "4"))
TASK2_CHUNK_SIZE = int(os.environ.get("TASK2_CHUNK_SIZE", "512"))
if any(value < 1 for value in (
    TASK2_BATCH_SIZE,
    TASK2_CPU_THREADS,

```

```

TASK2_MAX_SEQ_LENGTH,
TASK2_MODEL_WORKERS,
TASK2_BATCH_WORKERS,
TASK2_CHUNK_SIZE,
)):
    raise ValueError("Task 2 numeric settings must be positive")

if TASK2_PARALLEL_MODELS and TASK2_PARALLEL_BATCHES:
    print("[warning] Both parallel modes requested; using parallel batches only.
↪")
    TASK2_PARALLEL_MODELS = False
if TASK2_PARALLEL_MODELS and TASK2_DEVICE.lower() != "cpu":
    print("[warning] Parallel model encoding is enabled only for CPU; using
↪sequential encoding.")
    TASK2_PARALLEL_MODELS = False
if TASK2_PARALLEL_BATCHES and TASK2_DEVICE.lower() != "cpu":
    print("[warning] Parallel batch encoding is enabled only for CPU; using
↪sequential encoding.")
    TASK2_PARALLEL_BATCHES = False

_parallel_workers = min(TASK2_MODEL_WORKERS, len(MODEL_SPECS))
_batch_workers = min(TASK2_BATCH_WORKERS, max(1, len(corpus_texts)))
_parallel_enabled = TASK2_PARALLEL_MODELS and _parallel_workers > 1
_batch_parallel_enabled = TASK2_PARALLEL_BATCHES and _batch_workers > 1
_worker_count = _batch_workers if _batch_parallel_enabled else
↪_parallel_workers if _parallel_enabled else 1
_effective_threads = max(1, TASK2_CPU_THREADS // _worker_count)
if torch is not None and TASK2_DEVICE == "cpu":
    torch.set_num_threads(_effective_threads)
    try:
        torch.set_num_interop_threads(max(1, min(_effective_threads, 4)))
    except RuntimeError:
        pass

print(
    f"[Task 2] device={TASK2_DEVICE}, batch_size={TASK2_BATCH_SIZE}, "
    f"cpu_threads={_effective_threads}, max_seq_length={TASK2_MAX_SEQ_LENGTH}, "
    f"sort_by_length={TASK2_SORT_BY_LENGTH},
↪parallel_models={_parallel_enabled}, "
    f"parallel_batches={_batch_parallel_enabled}"
)

TIMINGS_PATH = "data/results/task2_timings.json"
try:
    timings = json.load(open(TIMINGS_PATH, encoding="utf-8")) if os.path.
↪exists(TIMINGS_PATH) else {}

```

```

except (OSError, json.JSONDecodeError):
    timings = {}

def _cache_is_usable(emb_path, expected_rows, expected_dim, expected_ids,
    ↪id_path):
    """Reject stale or incomplete arrays before attempting aligned loading."""
    if not (os.path.exists(emb_path) and os.path.exists(id_path)):
        return False
    try:
        shape = np.load(emb_path, mmap_mode="r").shape
        saved_ids = [str(x) for x in json.load(open(id_path, encoding="utf-8"))]
    except (OSError, ValueError, json.JSONDecodeError):
        return False
    return shape == (expected_rows, expected_dim) and set(saved_ids) ==
    ↪set(expected_ids)

def _encode_texts(model, texts, show_progress_bar, label, pool=None):
    """Encode length-sorted batches, then restore the original row order."""
    if TASK2_SORT_BY_LENGTH and len(texts) > 1:
        order = np.argsort([len(str(text)) for text in texts], kind="stable")
        texts_for_encoding = [texts[index] for index in order]
    else:
        order = None
        texts_for_encoding = texts

    print(f"    Encoding {label}: {len(texts)} texts...", flush=True)
    encode_kwargs = {
        "show_progress_bar": show_progress_bar,
        "batch_size": TASK2_BATCH_SIZE,
        "convert_to_numpy": True,
        "normalize_embeddings": False,
    }
    if pool is not None:
        encode_kwargs["pool"] = pool
        encode_kwargs["chunk_size"] = TASK2_CHUNK_SIZE
    embeddings = model.encode(texts_for_encoding, **encode_kwargs)
    if order is None:
        return embeddings

    restored = np.empty_like(embeddings)
    restored[order] = embeddings
    return restored

def _generate_model(model_name, spec):

```

```

"""Generate one model's arrays; this function is also used by CPU workers.
↪ """

from sentence_transformers import SentenceTransformer

model = None
pool = None
try:
    model = SentenceTransformer(resolve_model(model_name), ↵
↪ device=TASK2_DEVICE)
    model.max_seq_length = min(spec["max_input_length"], ↵
↪ TASK2_MAX_SEQ_LENGTH)
    show_progress_bar = not _parallel_enabled
    if _batch_parallel_enabled:
        print(
            f" {model_name}: starting {_batch_workers} batch workers "
            f"(chunk_size={TASK2_CHUNK_SIZE})...",
            flush=True,
        )
        pool = model.start_multi_process_pool(target_devices=["cpu"] * ↵
↪ _batch_workers)

    t0 = time.perf_counter()
    doc_emb = _encode_texts(model, corpus_texts, show_progress_bar, ↵
↪ "documents", pool)
    doc_elapsed = time.perf_counter() - t0
    print(f" Documents complete in {doc_elapsed:.1f}s", flush=True)

    t1 = time.perf_counter()
    qry_emb = _encode_texts(model, query_texts, show_progress_bar, ↵
↪ "queries", pool)
    qry_elapsed = time.perf_counter() - t1
    print(f" Queries complete in {qry_elapsed:.1f}s", flush=True)
    max_seq_length = model.max_seq_length
    return {
        "doc_emb": doc_emb,
        "qry_emb": qry_emb,
        "documents_s": doc_elapsed,
        "queries_s": qry_elapsed,
        "max_seq_length": max_seq_length,
    }
finally:
    if pool is not None:
        SentenceTransformer.stop_multi_process_pool(pool)
    if model is not None:
        del model
    gc.collect()

```

```

        if torch is not None and TASK2_DEVICE.lower().startswith("cuda"):
            torch.cuda.empty_cache()

model_records = {}
pending_models = []
for model_name, spec in MODEL_SPECS.items():
    print(f"=== {model_name} ===")
    safe = model_name.replace("/", "__")
    doc_path = f"data/results/embeddings_{safe}.npz"
    qry_path = f"data/results/query_embeddings_{safe}.npz"
    cache_ready = (
        _cache_is_usable(doc_path, len(doc_ids), spec["dim"], doc_ids, "data/
↳results/doc_ids.json")
        and _cache_is_usable(qry_path, len(query_ids), spec["dim"], query_ids,
↳"data/results/query_ids.json")
    )
    if cache_ready:
        doc_emb = load_aligned_embeddings(doc_path, "data/results/doc_ids.
↳json", doc_ids)
        qry_emb = load_aligned_embeddings(qry_path, "data/results/query_ids.
↳json", query_ids)
        saved_timing = timings.get(model_name)
        elapsed = saved_timing.get("documents_s") if isinstance(saved_timing,
↳dict) else saved_timing
        model_records[model_name] = {"doc_emb": doc_emb, "qry_emb": qry_emb,
↳"elapsed": elapsed}
        print(f"    Loaded saved embeddings: docs={doc_emb.shape},
↳queries={qry_emb.shape}")
    else:
        pending_models.append((model_name, spec))

def _record_generated_model(model_name, result):
    """Persist a completed result in the notebook's main thread."""
    if "error" in result:
        print(f"    [warning] Could not run this model here: {result['error']}")
        print("    Fix: install sentence-transformers + internet access, then
↳re-run.")
        model_records[model_name] = {"doc_emb": None, "qry_emb": None,
↳"elapsed": None}
        return

    doc_emb, qry_emb = result["doc_emb"], result["qry_emb"]
    safe = model_name.replace("/", "__")
    np.save(f"data/results/embeddings_{safe}.npz", doc_emb)

```

```

np.save(f"data/results/query_embeddings_{safe}.npy", qry_emb)
with open("data/results/doc_ids.json", "w", encoding="utf-8") as f:
    json.dump(doc_ids, f)
with open("data/results/query_ids.json", "w", encoding="utf-8") as f:
    json.dump(query_ids, f)

timings[model_name] = {
    "documents_s": round(result["documents_s"], 2),
    "queries_s": round(result["queries_s"], 2),
    "device": TASK2_DEVICE,
    "batch_size": TASK2_BATCH_SIZE,
    "cpu_threads": _effective_threads,
    "max_seq_length": result["max_seq_length"],
    "sort_by_length": TASK2_SORT_BY_LENGTH,
    "parallel_models": _parallel_enabled,
    "parallel_batches": _batch_parallel_enabled,
    "batch_workers": _batch_workers if _batch_parallel_enabled else 1,
    "chunk_size": TASK2_CHUNK_SIZE if _batch_parallel_enabled else None,
}
with open(TIMINGS_PATH, "w", encoding="utf-8") as f:
    json.dump(timings, f, indent=2)
model_records[model_name] = {
    "doc_emb": doc_emb,
    "qry_emb": qry_emb,
    "elapsed": result["documents_s"],
}
print(
    f" Generated + saved: docs={doc_emb.shape}, queries={qry_emb.shape}, "
    f"time={result['documents_s']:.1f}s"
)

if _parallel_enabled and len(pending_models) > 1:
    print(f"[Task 2] Encoding {len(pending_models)} models in parallel with_
↪{_parallel_workers} workers.")
    with ThreadPoolExecutor(max_workers=_parallel_workers) as executor:
        futures = {
            executor.submit(_generate_model, model_name, spec): model_name
            for model_name, spec in pending_models
        }
        for future in as_completed(futures):
            model_name = futures[future]
            try:
                result = future.result()
            except Exception as exc:
                result = {"error": repr(exc)}
            _record_generated_model(model_name, result)

```



```

else:
    for model_name, spec in pending_models:
        try:
            result = _generate_model(model_name, spec)
        except Exception as exc:
            result = {"error": repr(exc)}
        _record_generated_model(model_name, result)

comparison_rows = []
for model_name, spec in MODEL_SPECS.items():
    elapsed = model_records.get(model_name, {}).get("elapsed")
    comparison_rows.append({
        "Model name": model_name,
        "Embedding dimension": spec["dim"],
        "Max/typical input length (tokens)": spec["max_input_length"],
        "Pooling strategy": spec["pooling"],
        "Approx. document embedding time (s)": round(float(elapsed), 2) if
↪elapsed is not None else None,
        "Timing status": "Executed" if elapsed is not None else "Execution
↪required before submission",
    })

comparison_df = pd.DataFrame(comparison_rows)
comparison_df

```

```

[Task 2] device=cpu, batch_size=32, cpu_threads=12, max_seq_length=512,
sort_by_length=True, parallel_models=False, parallel_batches=False
=== distilbert-base-uncased ===
Loaded saved embeddings: docs=(5183, 768), queries=(300, 768)
=== BAAI/bge-large-en-v1.5 ===
Loaded saved embeddings: docs=(5183, 1024), queries=(300, 1024)

```

```

[54]:
      Model name  Embedding dimension \
0  distilbert-base-uncased           768
1   BAAI/bge-large-en-v1.5          1024

      Max/typical input length (tokens)  Pooling strategy \
0                                512      mean pooling
1                                512  [CLS] token pooling

      Approx. document embedding time (s)  Timing status
0                                341.12      Executed
1                                2034.55      Executed

```

3.1.1 Explanation of the Logic Used

Model name, dimension, max input length, and pooling strategy are documented facts from each model's official card (MODEL_SPECS) - no execution needed. Only the embeddings and timing require

real execution, so the loop attempts that per model and records whether execution completed and requires real timings before submission.

3.1.2 Performance and Parallel Encoding

The cell automatically uses CUDA when available and otherwise uses CPU inference with a bounded number of PyTorch threads. It sorts texts by length within each encode call and restores the original order, reducing padding work without changing the saved embedding alignment. Complete arrays are validated and reused from `data/results/`, so the expensive model pass runs only once.

The normal path shows the model's batch progress bar. For CPU experiments, set `TASK2_PARALLEL_BATCHES=1` to dispatch chunks of one model's corpus to a bounded Sentence Transformers multi-process pool. `TASK2_BATCH_WORKERS` controls the number of CPU worker processes and `TASK2_CHUNK_SIZE` controls how many texts are sent in each work item; `TASK2_BATCH_SIZE` still controls the inference mini-batch inside each worker. The parent process displays `Chunks` progress, and the pool is stopped after both document and query encoding.

Batch-level parallelism is deliberately opt-in because process startup, model memory, and inter-process transfer can make it slower than sequential CPU encoding on a laptop. It is mutually exclusive with `TASK2_PARALLEL_MODELS=1`; when both are requested, batch parallelism takes precedence. For a first CPU experiment, use `TASK2_PARALLEL_BATCHES=1`, `TASK2_BATCH_WORKERS=4`, and `TASK2_CHUNK_SIZE=512`, then compare the recorded timings rather than assuming a speedup. Parallel model encoding remains available with `TASK2_PARALLEL_MODELS=1`, but its two concurrent workers use phase messages instead of competing progress bars.

`TASK2_BATCH_SIZE`, `TASK2_CPU_THREADS`, `TASK2_DEVICE`, and `TASK2_MAX_SEQ_LENGTH` can also be overridden for a specific machine. The default sequence length remains 512 for the assignment comparison.

3.1.3 Justification for the Chosen Models

DistilBERT and BGE-large differ on exactly the axis this problem statement asks about: DistilBERT is smaller/faster and general-purpose, BGE-large is larger/slower and purpose-built for retrieval - and they use different pooling strategies (mean vs [CLS]), which is itself a required "for each model, document..." item. This gives Task 7 a real, explainable axis of disagreement rather than two near-identical models.

3.1.4 Inference

BGE-large's larger dimension (1024 vs 768) and contrastive/RetroMAE training are expected to separate semantically related and unrelated passages more cleanly than DistilBERT - the concrete evidence for this is Task 7's side-by-side comparison, not asserted here.

3.1.5 Limitations

- Embedding generation time is **hardware-dependent** (CPU vs GPU, memory bandwidth); the reported timings apply to the machine this notebook was run on and may not generalize.
- **distilbert-base-uncased** was not fine-tuned for retrieval; comparing it to a purpose-built retrieval model creates an **asymmetric comparison**. This is intentional for demonstrating the impact of training objective, but it does not represent a like-for-like architectural comparison.

- The corpus `text` field is used alone; the `title` field is discarded, potentially losing useful signal.

3.1.6 Possible Improvements

- Include query-specific encoding prefixes (e.g., BGE recommends prepending "**Represent this sentence:** " for certain use cases) to match the model's intended inference protocol.
- Benchmark with GPU encoding to obtain production-representative timing figures.
- Evaluate a third model (e.g., `sentence-transformers/all-mpnet-base-v2`) to broaden the comparison beyond two extremes.

4 Task 3 — Similarity Metric Comparison

Model used: `BAAI/bge-large-en-v1.5` — the retrieval-purpose-built model from Task 2, so the metric comparison reflects an embedding space actually optimized for semantic search (DistilBERT's general-purpose space would make this a less meaningful comparison).

Pair selection (20 pairs required): 5 randomly sampled queries $\times$ (all relevant documents from `qrels` + 4 random distractors each). SciFact queries typically have 1–2 relevant documents, giving 25 total pairs across 5 queries — enough per query to compare *rankings*, not just isolated scores.

```
[55]: # Purpose: implement the three similarity/distance metrics
# directly from their mathematical definitions (not by calling an opaque
# library function), so the formulas are visible and auditable.
import numpy as np

def cosine_similarity(a, b):
    """cos(theta) = (a . b) / (||a|| * ||b||)"""
    return float(np.dot(a, b) / (np.linalg.norm(a) * np.linalg.norm(b)))

def dot_product(a, b):
    """a . b = sum_i a_i * b_i"""
    return float(np.dot(a, b))

def l2_distance(a, b):
    """||a - b||_2 = sqrt(sum_i (a_i - b_i)^2)"""
    return float(np.linalg.norm(a - b))

print("[OK] Cosine, dot-product, and L2 formulas are defined for the metric_
↪experiment.")
```

[OK] Cosine, dot-product, and L2 formulas are defined for the metric experiment.

```
[56]: # Purpose: build the query-document pairs used for Task 3.
# For each of 5 randomly sampled queries, pair it with all its relevant
# documents (from qrels) plus 4 random "distractor" documents -- so each
```

```

# query has a small ranked set of candidates to compare metrics across.
import random
random.seed(129)

MODEL_FOR_TASK3 = "BAAI/bge-large-en-v1.5"
N_QUERIES, N_DISTRACTORS = 5, 4

sample_queries = queries_df.sample(n=N_QUERIES, random_state=129).
↳reset_index(drop=True)
all_doc_ids = corpus_df["doc_id"].tolist()

pairs = []
for _, q in sample_queries.iterrows():
    qid, qtext = q["query_id"], q["text"]

    # Include all known relevant documents and sample only non-relevant_
    ↳distractors.
    relevant_ids = set(
        qrels_df.loc[qrels_df["query_id"] == qid, "doc_id"].astype(str)
    )
    distractor_pool = [d for d in all_doc_ids if d not in relevant_ids]
    distractors = random.sample(
        distractor_pool, min(N_DISTRACTORS, len(distractor_pool))
    )
    doc_ids_for_query = list(relevant_ids) + distractors

    for did in doc_ids_for_query:
        text = corpus_df.loc[corpus_df["doc_id"] == did, "text"].values[0]
        pairs.append({"query_id": qid, "query_text": qtext, "doc_id": did,
                      "doc_text": text, "is_relevant": did in relevant_ids})

pairs_df = pd.DataFrame(pairs)
print(f"Selected {len(pairs_df)} query-document pairs across_
↳{pairs_df['query_id'].nunique()} "
      f"queries (>= 20 required).")

```

Selected 25 query-document pairs across 5 queries (>= 20 required).

```

[57]: # Purpose: get an embedding for each pair's query/document, then
# compute cosine, dot product, and L2 for every pair. Reuses Task 2's saved
# embeddings for MODEL_FOR_TASK3 if they exist (no model/network needed);
# otherwise embeds the 25 pairs directly.
import os
import numpy as np
import pandas as pd

_safe = MODEL_FOR_TASK3.replace("/", "__")

```

```

_doc_path = f"data/results/embeddings_{_safe}.npz"
_qry_path = f"data/results/query_embeddings_{_safe}.npz"
metrics_df = q_emb = d_emb = None

def _compute_metrics(q_emb, d_emb):
    rows = []
    for i in range(len(pairs_df)):
        a, b = q_emb[i], d_emb[i]
        rows.append({
            "query_id": pairs_df.loc[i, "query_id"],
            "doc_id": pairs_df.loc[i, "doc_id"],
            "is_relevant": pairs_df.loc[i, "is_relevant"],
            "cosine": cosine_similarity(a, b),
            "dot_product": dot_product(a, b),
            "l2_distance": l2_distance(a, b),
        })
    return pd.DataFrame(rows)

if os.path.exists(_doc_path) and os.path.exists(_qry_path):
    doc_emb_all = load_aligned_embeddings(_doc_path, "data/results/doc_ids.
↪json", corpus_df["doc_id"])
    qry_emb_all = load_aligned_embeddings(_qry_path, "data/results/query_ids.
↪json", queries_df["query_id"])
    doc_idx = {d: i for i, d in enumerate(corpus_df["doc_id"].tolist())}
    qry_idx = {q: i for i, q in enumerate(queries_df["query_id"].tolist())}
    q_emb = np.stack([qry_emb_all[qry_idx[pairs_df.loc[i, "query_id"]]] for i in
↪range(len(pairs_df))])
    d_emb = np.stack([doc_emb_all[doc_idx[pairs_df.loc[i, "doc_id"]]] for i in
↪range(len(pairs_df))])
    metrics_df = _compute_metrics(q_emb, d_emb)
    print(f"Reused saved {MODEL_FOR_TASK3} embeddings for {len(metrics_df)}
↪pairs.")
    display(metrics_df)
else:
    try:
        from sentence_transformers import SentenceTransformer
        model = SentenceTransformer(resolve_model(MODEL_FOR_TASK3))
        q_emb = model.encode(pairs_df["query_text"].tolist(),
↪show_progress_bar=True)
        d_emb = model.encode(pairs_df["doc_text"].tolist(),
↪show_progress_bar=True)
        metrics_df = _compute_metrics(q_emb, d_emb)
        print(f"Computed cosine/dot/L2 for {len(metrics_df)} pairs using
↪{MODEL_FOR_TASK3}.")

```

```

display(metrics_df)
except Exception as e:
    print(f"[warning] Could not run {MODEL_FOR_TASK3} here: {e!r}")
    print("Run Task 2 first to create saved embeddings, then re-run this_
↪cell.")

```

Reused saved BAAI/bge-large-en-v1.5 embeddings for 25 pairs.

	query_id	doc_id	is_relevant	cosine	dot_product	l2_distance
0	660	1215116	True	0.613863	0.613863	0.878792
1	660	41496215	False	0.407035	0.407035	1.089004
2	660	32177659	False	0.447672	0.447672	1.051026
3	660	13702924	False	0.500359	0.500359	0.999641
4	660	2048139	False	0.525342	0.525342	0.974329
5	1362	8290953	True	0.584091	0.584090	0.912041
6	1362	22025252	False	0.489939	0.489939	1.010011
7	1362	154763124	False	0.441601	0.441601	1.056787
8	1362	10795340	False	0.477080	0.477080	1.022663
9	1362	17119869	False	0.498330	0.498330	1.001669
10	185	18340282	True	0.674860	0.674860	0.806399
11	185	11335860	False	0.531729	0.531729	0.967751
12	185	4702639	False	0.547632	0.547632	0.951176
13	185	36288526	False	0.444918	0.444918	1.053644
14	185	6826636	False	0.481996	0.481996	1.017845
15	743	32159283	True	0.776836	0.776836	0.668078
16	743	6082738	False	0.517617	0.517617	0.982225
17	743	13798951	False	0.510560	0.510560	0.989383
18	743	38784540	False	0.501756	0.501756	0.998242
19	743	11291348	False	0.530083	0.530083	0.969450
20	1359	11614737	True	0.866433	0.866433	0.516850
21	1359	6144969	False	0.450090	0.450090	1.048723
22	1359	27024392	False	0.553945	0.553945	0.944516
23	1359	43192375	False	0.468878	0.468878	1.030652
24	1359	25440070	False	0.540895	0.540895	0.958233

```

[58]: # Purpose: two checks required by Task 3.
# 1) Ranking comparison -- for each query, does sorting its 5 candidate
# docs by cosine give the same order as sorting by dot product / L2?
# Also prints the raw embedding norms, since dot product only diverges
# from cosine when vector norms actually vary across documents.
# 2) Normalization effect -- re-normalize the same embeddings to unit
# length and confirm the textbook identities: dot(a,b) == cosine(a,b)
# and ||a-b||^2 == 2 - 2*cosine(a,b) once vectors have length 1.
if metrics_df is not None:
    doc_norms = np.linalg.norm(d_emb, axis=1)
    print(f"Raw document embedding norms: mean={doc_norms.mean():.4f},
↪std={doc_norms.std():.4f} "

```

```

        f"(dot product can only disagree with cosine if this std is_
↪non-trivial)\n")

    print("=== Ranking comparison per query (raw, unnormalized embeddings) ===")
    n_changed = 0
    for qid, group in metrics_df.groupby("query_id"):
        cos_rank = group.sort_values("cosine", ascending=False)["doc_id"].
↪tolist()
        dot_rank = group.sort_values("dot_product", ascending=False)["doc_id"].
↪tolist()
        l2_rank = group.sort_values("l2_distance", ascending=True)["doc_id"].
↪tolist() # smaller distance = closer
        changed = not (cos_rank == dot_rank == l2_rank)
        n_changed += changed
        print(f"query {qid}: cosine==dot ranking? {cos_rank == dot_rank} | "
              f"cosine==L2 ranking? {cos_rank == l2_rank}")
    print(f"Summary: ranking changed across metrics for {n_changed}/
↪{metrics_df['query_id'].nunique()} queries.\n")

    print("=== Normalization effect: recompute dot/L2 on unit-normalized_
↪embeddings ===")
    def normalize(v):
        return v / np.linalg.norm(v)

    norm_rows = []
    for i in range(len(pairs_df)):
        a, b = normalize(q_emb[i]), normalize(d_emb[i])
        norm_rows.append({
            "query_id": pairs_df.loc[i, "query_id"],
            "doc_id": pairs_df.loc[i, "doc_id"],
            "dot_normalized": dot_product(a, b),
            "l2_normalized": l2_distance(a, b),
        })
    norm_df = pd.DataFrame(norm_rows).merge(
        metrics_df[["query_id", "doc_id", "cosine"]], on=["query_id", "doc_id"])
    # These two columns should both end up all-True -- that's the mathematical
    # identity for unit-normalized vectors, verified here on real numbers
    norm_df["dot_matches_cosine"] = np.isclose(norm_df["dot_normalized"],
↪norm_df["cosine"], atol=1e-4)
    norm_df["l2sq_matches_2_minus_2cos"] = np.isclose(
        norm_df["l2_normalized"] ** 2, 2 - 2 * norm_df["cosine"], atol=1e-4)
    display(norm_df[["query_id", "doc_id", "dot_matches_cosine",
↪"l2sq_matches_2_minus_2cos"]])
    print(f"Summary: identity held for {norm_df['dot_matches_cosine'].sum()}/
↪{len(norm_df)} pairs (dot) ")

```

```

        f"and {norm_df['l2sq_matches_2_minus_2cos'].sum()}/{len(norm_df)}_
pairs (L2).")
else:
    print("Skipped -- needs metrics_df from the cell above (requires live model_
access).")

```

Raw document embedding norms: mean=1.0000, std=0.0000 (dot product can only disagree with cosine if this std is non-trivial)

```

=== Ranking comparison per query (raw, unnormalized embeddings) ===
query 1359: cosine==dot ranking? True | cosine==L2 ranking? True
query 1362: cosine==dot ranking? True | cosine==L2 ranking? True
query 185: cosine==dot ranking? True | cosine==L2 ranking? True
query 660: cosine==dot ranking? True | cosine==L2 ranking? True
query 743: cosine==dot ranking? True | cosine==L2 ranking? True
Summary: ranking changed across metrics for 0/5 queries.

```

=== Normalization effect: recompute dot/L2 on unit-normalized embeddings ===

	query_id	doc_id	dot_matches_cosine	l2sq_matches_2_minus_2cos
0	660	1215116	True	True
1	660	41496215	True	True
2	660	32177659	True	True
3	660	13702924	True	True
4	660	2048139	True	True
5	1362	8290953	True	True
6	1362	22025252	True	True
7	1362	154763124	True	True
8	1362	10795340	True	True
9	1362	17119869	True	True
10	185	18340282	True	True
11	185	11335860	True	True
12	185	4702639	True	True
13	185	36288526	True	True
14	185	6826636	True	True
15	743	32159283	True	True
16	743	6082738	True	True
17	743	13798951	True	True
18	743	38784540	True	True
19	743	11291348	True	True
20	1359	11614737	True	True
21	1359	6144969	True	True
22	1359	27024392	True	True
23	1359	43192375	True	True
24	1359	25440070	True	True

Summary: identity held for 25/25 pairs (dot) and 25/25 pairs (L2).


```
[59]: # Purpose: normalization experiment on genuinely UN-normalized vectors.
# BGE-large's embeddings come out already unit-length, so normalization has no
# visible effect there (a valid finding, but nothing to observe). DistilBERT's
# mean-pooled embeddings are NOT normalized (their norms vary), so the SAME 25
# pairs reveal a real divergence between dot-product and cosine rankings that
# normalization then removes -- demonstrating exactly WHEN normalization
↳ matters.
import os
import numpy as np

_DEMO = "distilbert-base-uncased"
_dsafe = _DEMO.replace("/", "__")
_ddoc = f"data/results/embeddings_{_dsafe}.npz"
_dqry = f"data/results/query_embeddings_{_dsafe}.npz"

if metrics_df is not None and os.path.exists(_ddoc) and os.path.exists(_dqry):
    _doc_all = load_aligned_embeddings(_ddoc, "data/results/doc_ids.json",
↳ corpus_df["doc_id"])
    _qry_all = load_aligned_embeddings(_dqry, "data/results/query_ids.json",
↳ queries_df["query_id"])
    _didx = {d: i for i, d in enumerate(corpus_df["doc_id"].tolist())}
    _qidx = {q: i for i, q in enumerate(queries_df["query_id"].tolist())}
    dq = np.stack([_qry_all[_qidx[pairs_df.loc[i, "query_id"]]] for i in
↳ range(len(pairs_df))])
    dd = np.stack([_doc_all[_didx[pairs_df.loc[i, "doc_id"]]] for i in
↳ range(len(pairs_df))])

    dn = np.linalg.norm(dd, axis=1)
    print(f"{_DEMO}: raw document-vector norms mean={dn.mean():.3f}, std={dn.
↳ std():.3f} "
          f"(range {dn.min():.2f}-{dn.max():.2f}) -- genuinely un-normalized.
↳ \n")

    demo = pd.DataFrame([
        "query_id": pairs_df.loc[i, "query_id"],
        "doc_id": pairs_df.loc[i, "doc_id"],
        "cosine": cosine_similarity(dq[i], dd[i]),
        "dot_product": dot_product(dq[i], dd[i]),
        "l2_distance": l2_distance(dq[i], dd[i]),
    ] for i in range(len(pairs_df)))

    print("=== RAW (un-normalized) ranking comparison per query ===")
    n_diff = 0
    for qid, g in demo.groupby("query_id"):
        cr = g.sort_values("cosine", ascending=False)["doc_id"].tolist()
        dr = g.sort_values("dot_product", ascending=False)["doc_id"].tolist()
```

```

    lr = g.sort_values("l2_distance")["doc_id"].tolist()
    n_diff += (cr != dr)
    print(f"query {qid}: cosine==dot ranking? {cr == dr} | cosine==L2_
↳ranking? {cr == lr}")
    print(f"Summary: dot-product ranking DIFFERS from cosine for {n_diff}/
↳{demo['query_id'].nunique()} "
          f"queries -- purely because the raw vector norms vary.\n")

    print("=== After L2-normalization (unit vectors) ===")
    okd = okl = 0
    for i in range(len(pairs_df)):
        a = dq[i] / np.linalg.norm(dq[i]); b = dd[i] / np.linalg.norm(dd[i])
        okd += bool(np.isclose(dot_product(a, b), demo.loc[i, "cosine"],
↳atol=1e-4))
        okl += bool(np.isclose(l2_distance(a, b) ** 2, 2 - 2 * demo.loc[i,
↳"cosine"], atol=1e-4))
    print(f"dot(a,b) == cosine(a,b) held for {okd}/{len(pairs_df)} pairs")
    print(f"||a-b||^2 == 2 - 2*cosine held for {okl}/{len(pairs_df)} pairs")
    print("\nConclusion: when norms vary (DistilBERT) normalization changes the_
↳dot-product "
          "ranking to agree with cosine; when vectors are already unit-length_
↳(BGE) it has no "
          "effect. This is the practical reason retrieval systems normalize_
↳before using dot product.")
else:
    print("Skipped -- needs metrics_df (cell above) and DistilBERT embeddings_
↳from Task 2.")

```

distilbert-base-uncased: raw document-vector norms mean=7.851, std=0.313 (range 7.42-8.56) -- genuinely un-normalized.

=== RAW (un-normalized) ranking comparison per query ===

```

query 1359: cosine==dot ranking? False | cosine==L2 ranking? True
query 1362: cosine==dot ranking? False | cosine==L2 ranking? False
query 185: cosine==dot ranking? False | cosine==L2 ranking? True
query 660: cosine==dot ranking? True | cosine==L2 ranking? False
query 743: cosine==dot ranking? False | cosine==L2 ranking? False
Summary: dot-product ranking DIFFERS from cosine for 4/5 queries -- purely
because the raw vector norms vary.

```

=== After L2-normalization (unit vectors) ===

```

dot(a,b) == cosine(a,b) held for 25/25 pairs
||a-b||^2 == 2 - 2*cosine held for 25/25 pairs

```

Conclusion: when norms vary (DistilBERT) normalization changes the dot-product ranking to agree with cosine; when vectors are already unit-length (BGE) it has no effect. This is the practical reason retrieval systems normalize before using

dot product.

4.0.1 Explanation of the Logic Used

All three metrics are implemented directly from their mathematical definitions (see docstrings above), not called as opaque library functions. The sampled pairs let rankings be compared *per query* across all three metrics, and the normalization cells re-embed nothing — they unit-normalize the same vectors and recompute, isolating normalization as the only variable. Raw embedding-norm statistics are printed alongside each ranking comparison, since dot product can only diverge from cosine when document norms actually vary.

4.0.2 Justification for the Chosen Approach

Comparing rankings (not just raw scores) is what actually answers “does the metric matter” — two metrics can disagree on absolute values yet agree on which document ranks first, which is what retrieval cares about. Testing the normalized case directly demonstrates the identities $\text{dot}(\hat{\mathbf{a}}, \hat{\mathbf{b}}) = \cos(\mathbf{a}, \mathbf{b})$ and $\|\hat{\mathbf{a}} - \hat{\mathbf{b}}\|^2 = 2 - 2 \cdot \cos(\mathbf{a}, \mathbf{b})$ for unit vectors $\hat{\mathbf{a}}, \hat{\mathbf{b}}$ — shown to hold on the actual embeddings, not asserted from theory.

4.0.3 The normalization experiment (two models, on purpose)

The selected model, **BGE-large**, emits already unit-length vectors (sentence-transformers L2-normalizes inside `encode()` for this model): norm mean=1.0000, std=0.0000. So for BGE the dot-product, cosine, and L2 rankings are identical for **0/5** queries changed — normalization has nothing left to change. That is itself a valid finding, but it shows no *effect*.

To observe the effect, the same pairs are re-scored with **DistilBERT’s mean-pooled embeddings, which are genuinely un-normalized** (document-norm mean 7.85, std 0.31, range 7.4–8.6). There the result is the opposite:

	Dot-product vs cosine ranking
Raw (un-normalized) DistilBERT	differs for 4/5 queries — norms vary, so dot product rewards longer vectors
After L2-normalization	dot == cosine for all pairs ; $\ \mathbf{a} - \mathbf{b}\ ^2 == 2 - 2 \cdot \cos$ for all pairs

4.0.4 Inference

When vector norms vary, the metric choice **changes the ranking**: dot product favours high-norm documents, while cosine ignores magnitude. Normalizing to unit length collapses the three metrics onto the same ordering (dot becomes cosine; ascending L2 matches descending cosine). When vectors are already unit-length (BGE), the choice is irrelevant. This is the concrete, executed reason production retrieval systems normalize embeddings before scoring with dot product — verified numerically here rather than argued from theory alone.

4.0.5 Limitations

- The experiment uses only **5 queries and a small candidate set per query**; ranking differences on a full corpus with 5,183 documents could behave differently.

- The **lexical overlap metric** (`word_overlap`) **does not strip punctuation** ("cancer." and "cancer" are treated as different tokens), which causes systematic underestimation of literal word overlap.
- Metric comparison is done on a single model's embedding space (BGE-large); conclusions about which metric is best may differ for other embedding spaces.

4.0.6 Possible Improvements

- Extend the ranking comparison to all 300 queries and the full 5,183-document candidate set.
- Normalise punctuation before computing lexical overlap (use `re.sub(r'[\w\s]', '', text)` before splitting).
- Include Spearman's rank correlation as a more robust ranking-agreement metric than pairwise equality.

5 Task 4 — Exact kNN Baseline

Model used: BAAI/bge-large-en-v1.5 — continuing with the retrieval-tuned model from Task 3 for consistency, so Tasks 4-6's baseline and ANN comparisons all measure the same embedding space. Task 7 brings DistilBERT back in for the cross-model comparison.

Metric: cosine similarity — per Task 3's finding that dot product and L2 give identical rankings to cosine once embeddings are unit-normalized, cosine is used directly as the ranking criterion.

```
[60]: # Purpose: load the full BGE-large corpus + query embeddings
# (reusing Task 2's saved arrays, or generating them if Task 2 wasn't run for
# real yet), aligned to corpus_df / queries_df row order.
import os
import numpy as np

BASELINE_MODEL = "BAAI/bge-large-en-v1.5"
_safe = BASELINE_MODEL.replace("/", "_")
_doc_path = f"data/results/embeddings_{_safe}.npz"
_qry_path = f"data/results/query_embeddings_{_safe}.npz"

doc_emb = qry_emb = None
if os.path.exists(_doc_path) and os.path.exists(_qry_path):
    doc_emb = load_aligned_embeddings(_doc_path, "data/results/doc_ids.json",
    ↪corpus_df["doc_id"])
    qry_emb = load_aligned_embeddings(_qry_path, "data/results/query_ids.json",
    ↪queries_df["query_id"])
    print(f"Loaded saved embeddings: docs={doc_emb.shape}, queries={qry_emb.
    ↪shape}")
else:
    try:
        from sentence_transformers import SentenceTransformer
        model = SentenceTransformer(resolve_model(BASELINE_MODEL))
```

```

        doc_emb = model.encode(corpus_df["text"].tolist(),
↪show_progress_bar=True, batch_size=64)
        qry_emb = model.encode(queries_df["text"].tolist(),
↪show_progress_bar=True, batch_size=64)
        print(f"Generated embeddings: docs={doc_emb.shape}, queries={qry_emb.
↪shape}")
        except Exception as e:
            print(f"[warning] Could not load or generate {BASELINE_MODEL}
↪embeddings: {e!r}")
            print("Run Task 2 first (needs sentence-transformers + internet), then
↪re-run this cell.")

# Unit-normalize once -- makes cosine similarity a plain dot product (Task 3's
↪identity)
if doc_emb is not None:
    doc_emb_norm = doc_emb / np.linalg.norm(doc_emb, axis=1, keepdims=True)
    qry_emb_norm = qry_emb / np.linalg.norm(qry_emb, axis=1, keepdims=True)

```

Loaded saved embeddings: docs=(5183, 1024), queries=(300, 1024)

```

[61]: # Purpose: exact brute-force kNN. For every query, score it against ALL
# corpus documents (cosine == dot product on the normalized vectors), take the
# top 5, and measure Recall@5 (vs qrels) plus per-query latency. Latency is the
# MEDIAN of several timed runs (time.perf_counter) taken after a warm-up, so the
# number is stable instead of being dominated by one-off timer noise -- and it
# is measured the same way as the ANN indexes in Task 5, keeping the speedup
# comparison fair.
import time
import numpy as np
import pandas as pd

TOP_K = 5
LAT_REPEATS = 25          # timed runs per query; the median is reported
results = []

if doc_emb is not None:
    doc_ids = corpus_df["doc_id"].tolist()

    # Warm-up: a few untimed searches so BLAS threads / caches are hot; the
    # very first calls are otherwise misleadingly slow and inflate latency.
    for _ in range(5):
        _ = np.argsort(-(doc_emb_norm @ qry_emb_norm[0]))[:TOP_K]

    for i, row in queries_df.iterrows():
        qid = row["query_id"]
        q_vec = qry_emb_norm[i]

```

```

# Result + correctness (computed once).
scores = doc_emb_norm @ q_vec
top_idx = np.argsort(-scores)[:TOP_K]
retrieved_ids = [doc_ids[j] for j in top_idx]
relevant_ids = set(qrels_df.loc[qrels_df["query_id"] == qid, "doc_id"])
hits = len(set(retrieved_ids) & relevant_ids)
recall_at_5 = hits / len(relevant_ids) if relevant_ids else 0.0

# Latency: median of LAT_REPEATS timed runs of the same exact search.
times = np.empty(LAT_REPEATS)
for r in range(LAT_REPEATS):
    t0 = time.perf_counter()
    s = doc_emb_norm @ q_vec
    _ = np.argsort(-s)[:TOP_K]
    times[r] = time.perf_counter() - t0
latency = float(np.median(times))

results.append({
    "query_id": qid,
    "latency_s": latency,
    "recall_at_5": recall_at_5,
    "vectors_examined": len(doc_ids), # exact search always checks
    "retrieved_doc_ids": retrieved_ids,
})

knn_results_df = pd.DataFrame(results)

avg_latency = float(knn_results_df["latency_s"].mean()) # feeds ANN
med_latency = float(knn_results_df["latency_s"].median())
avg_recall = knn_results_df["recall_at_5"].mean()
print(f"Queries evaluated      : {len(knn_results_df)}")
print(f"Latency/query (median): {med_latency*1000:.3f} ms      "
      f"(mean {avg_latency*1000:.3f} ms; median of {LAT_REPEATS} timed runs
each, perf_counter)")
print(f"Average Recall@5      : {avg_recall:.3f}")
print(f"Vectors examined       : {len(doc_ids)} per query (exact search
scans the whole corpus)")
else:
    knn_results_df = None
    avg_latency = None
    print("Skipped -- needs doc_emb/qry_emb from the cell above.")

```

```

Queries evaluated      : 300
Latency/query (median): 0.231 ms      (mean 0.235 ms; median of 25 timed runs each,
perf_counter)

```

Average Recall@5 : 0.780
Vectors examined : 5183 per query (exact search scans the whole corpus)

5.0.1 Explanation of the Logic Used

Cosine similarity is one matrix-vector dot product against every document (Task 3 showed dot product on normalized vectors equals cosine). `argsort` picks the top 5. “Vectors examined” is the full corpus size — exact search has no shortcut.

5.0.2 Justification for the Chosen Approach

A genuinely exhaustive baseline is required so Task 5/6’s “speed relative to exact search” comparison means something real.

5.0.3 Inference

The executed output reports the measured mean latency, Recall@5, and full-corpus vector count. Latency is hardware-dependent; Recall@5 is interpreted together with the incomplete/binary qrels rather than as a claim that every retrieved result is incorrect.

5.0.4 Limitations

- **Latency is hardware-dependent:** median of 25 timed runs stabilizes the measurement, but numbers are specific to the CPU/memory configuration used.
- **Binary, incomplete qrels:** a retrieved document that is relevant but has no qrel entry is counted as a miss, so Recall@5 is a lower-bound estimate of true recall.
- The exact search **does not use BLAS-level batching** across queries (one query at a time), so the per-query latency may be higher than a batched production system.

5.0.5 Possible Improvements

- Report **Precision@5** alongside Recall@5 to characterize retrieval from both directions.
- Batch all query vectors into a single matrix multiplication (`qry_emb_norm @ doc_emb_norm.T`) to get production-representative throughput.
- Compare latency across hardware platforms (CPU vs GPU) to show how the relative advantage of exact vs ANN shifts.

6 Task 5 — HNSW vs IVF ANN Search

Both indexes are built on the same BGE-large embeddings and evaluated against the same 300 queries used in Task 4, so results are directly comparable to the exact baseline.

- **HNSW** (`faiss.IndexHNSWFlat`): the search-time parameter varied is `efSearch` — the size of the candidate list explored during the graph search (higher = more accurate, slower).
- **IVF** (`faiss.IndexIVFFlat`): `nprobe` — the number of index clusters (cells) checked per query — is varied across 4 values (3 required).

Both indexes use inner product on the unit-normalized embeddings, which is equivalent to cosine similarity (per Task 3’s identity) and matches Task 4’s exact-search metric.

```

[62]: # Purpose: build one HNSW index and one IVF index on the same
# normalized BGE-large document embeddings used in Task 4, plus a helper that
# runs a query set against any FAISS index and reports Recall@5 (vs qrels
# ground truth, same definition as Task 4), latency, and speedup vs exact
↪search.
import time
import numpy as np
import pandas as pd

try:
    import faiss
    FAISS_OK = True
except Exception as e:
    faiss = None
    FAISS_OK = False
    print(f"[warning] faiss not available here: {e!r}")
    print("Install with: pip install faiss-cpu, then re-run this cell.")

hnsf_index = ivf_index = None
if FAISS_OK and doc_emb is not None:
    dim = doc_emb_norm.shape[1]
    doc_vecs = np.ascontiguousarray(doc_emb_norm.astype("float32"))

    # HNSW -- M=32 is a reasonable build-time default; efSearch is varied later
    hnsf_index = faiss.IndexHNSWFlat(dim, 32, faiss.METRIC_INNER_PRODUCT)
    hnsf_index.add(doc_vecs)

    # IVF -- nlist=100 clusters is reasonable for a ~5k-document corpus
    NLIST = 100
    quantizer = faiss.IndexFlatIP(dim)
    ivf_index = faiss.IndexIVFFlat(quantizer, dim, NLIST, faiss.
↪METRIC_INNER_PRODUCT)
    ivf_index.train(doc_vecs)
    ivf_index.add(doc_vecs)

    print(f"Built HNSW (M=32) and IVF (nlist={NLIST}) indexes on
↪{len(doc_vecs)} vectors.")

LAT_REPEATS = 25 # timed runs per query; median reported (matches Task 4)

def evaluate_index(index, label, param_name, param_value, exact_avg_latency):
    """Run all queries through `index`, return one summary row: Recall@5
    (against qrels), average latency, and speedup vs exact search. Latency per
    query is the MEDIAN of LAT_REPEATS timed runs (time.perf_counter) after a
    warm-up -- the same protocol used for the exact baseline in Task 4, so the

```



```

speedup ratio compares like with like."""
doc_ids = corpus_df["doc_id"].tolist()
latencies, recalls = [], []

# Warm-up: a few untimed searches so the index's internal caches are hot.
for _ in range(5):
    index.search(qry_emb_norm[0:1].astype("float32"), 5)

for i, row in queries_df.iterrows():
    qid = row["query_id"]
    q_vec = qry_emb_norm[i:i+1].astype("float32")

    # Result + correctness (computed once).
    _, top_idx = index.search(q_vec, 5)
    retrieved_ids = [doc_ids[j] for j in top_idx[0] if j != -1]
    relevant_ids = set(qrels_df.loc[qrels_df["query_id"] == qid, "doc_id"])
    hits = len(set(retrieved_ids) & relevant_ids)
    recalls.append(hits / len(relevant_ids) if relevant_ids else 0.0)

    # Latency: median of LAT_REPEATS timed runs of the same index search.
    times = np.empty(LAT_REPEATS)
    for r in range(LAT_REPEATS):
        t0 = time.perf_counter()
        index.search(q_vec, 5)
        times[r] = time.perf_counter() - t0
    latencies.append(float(np.median(times)))

avg_latency = float(np.mean(latencies))
return {
    "method": label,
    "param_name": param_name,
    "param_value": param_value,
    "recall_at_5": float(np.mean(recalls)),
    "avg_latency_s": avg_latency,
    "speedup_vs_exact": (exact_avg_latency / avg_latency)
                        if (exact_avg_latency is not None and avg_latency_
↪ > 0) else None,
}

```

Built HNSW (M=32) and IVF (nlist=100) indexes on 5183 vectors.

```

[63]: # Purpose: sweep efSearch (HNSW's search-time recall/speed knob)
# across 4 values and evaluate each configuration.
ann_rows = []
ann_results_df = None
if hnsw_index is not None:
    for ef in (16, 32, 64, 128):

```

```

hnsw_index.hnswefSearch = ef
row = evaluate_index(hnsw_index, "HNSW", "efSearch", ef, avg_latency)
row["candidate_vectors_estimate"] = ef # efSearch is a search-effort,
↳proxy, not an exact count
ann_rows.append(row)
print(f"HNSW efSearch={ef:>3}: Recall@5={row['recall_at_5']:.3f}, "
      f"latency={row['avg_latency_s']*1000:.3f}ms,
↳speedup={row['speedup_vs_exact'] if row['speedup_vs_exact'] is None else
↳round(row['speedup_vs_exact'],2)}x")
else:
    print("Skipped -- needs hnsw_index from the cell above (requires faiss-cpu).
↳")

```

```

HNSW efSearch= 16: Recall@5=0.773, latency=0.036ms, speedup=6.52x
HNSW efSearch= 32: Recall@5=0.780, latency=0.060ms, speedup=3.94x
HNSW efSearch= 64: Recall@5=0.780, latency=0.115ms, speedup=2.04x
HNSW efSearch=128: Recall@5=0.780, latency=0.207ms, speedup=1.13x

```

```

[64]: # Purpose: sweep nprobe (IVF's search-time recall/speed knob,
# number of clusters checked per query) across 4 values and evaluate each.
if ivf_index is not None:
    NLIST = ivf_index.nlist
    for nprobe in (1, 5, 10, 20):
        ivf_index.nprobe = nprobe
        row = evaluate_index(ivf_index, "IVF", "nprobe", nprobe, avg_latency)
        # Fraction of clusters probed * total vectors approximates candidates
↳examined
        row["candidate_vectors_estimate"] = round(nprobe / NLIST *
↳len(corpus_df))
        ann_rows.append(row)
        print(f"IVF nprobe={nprobe:>3}: Recall@5={row['recall_at_5']:.3f}, "
              f"latency={row['avg_latency_s']*1000:.3f}ms,
↳speedup={row['speedup_vs_exact'] if row['speedup_vs_exact'] is None else
↳round(row['speedup_vs_exact'],2)}x")

    ann_results_df = pd.DataFrame(ann_rows)
    ann_results_df
else:
    ann_results_df = None
    print("Skipped -- needs ivf_index from the cell above (requires faiss-cpu).
↳")

```

```

IVF nprobe= 1: Recall@5=0.502, latency=0.015ms, speedup=15.52x
IVF nprobe= 5: Recall@5=0.716, latency=0.035ms, speedup=6.7x
IVF nprobe= 10: Recall@5=0.736, latency=0.064ms, speedup=3.66x
IVF nprobe= 20: Recall@5=0.769, latency=0.133ms, speedup=1.76x

```

6.0.1 Explanation of the Logic Used

`evaluate_index()` reuses Task 4’s `Recall@5` definition and query loop, swapping in `index.search()`. `candidate_vectors_estimate` is an estimate for IVF and a search-effort proxy for HNSW (FAISS doesn’t expose the true count). IVF uses `nlist=100` clusters (slightly above the $\sqrt{N}$ 72 rule of thumb, giving finer cluster granularity at negligible extra build cost for 5k vectors).

6.0.2 Justification for the Chosen Approach

`efSearch` and `nprobe` are each index’s standard recall/speed knob — sweeping them traces the trade-off curve Task 6 needs.

6.0.3 Inference

The executed table and plots are used to identify the fastest, highest-recall, and balanced configurations. Increasing `efSearch` or `nprobe` generally expands the search effort, but the measured values—not an assumed monotonic pattern—determine the recommendation.

6.0.4 Limitations

- **Index build time and memory consumption** are not measured; HNSW builds slowly and has a large graph memory footprint compared to IVF — this cost is invisible in these results.
- `candidate_vectors_estimate` for HNSW is a **proxy** (`efSearch` value), not a true count, since FAISS does not expose the exact number of distance computations per query.
- The speedup is measured on a **5k-document corpus**; HNSW and IVF scale differently, and relative speedups at 1M+ documents would differ substantially.

6.0.5 Possible Improvements

- Measure index build time and peak RAM for both HNSW and IVF to give a complete cost picture.
- Test broader parameter ranges (HNSW `M=16` vs `M=32` vs `M=64`; IVF `nlist` from 50 to 200) to better characterize the build vs search trade-off.
- Add Product Quantization (IVFPQ) as a third ANN variant to demonstrate compression-based speed-ups.

7 Task 6 — ANN Trade-off Analysis

Uses `ann_results_df` from Task 5 (8 configurations: 4 HNSW `efSearch` values + 4 IVF `nprobe` values) to plot the recall/latency/efficiency trade-off and pick a configuration for production.

```
[65]: # Task 6 plots: show the measured recall/latency trade-off clearly.
import os
import matplotlib.pyplot as plt

if ann_results_df is not None and len(ann_results_df):
```

```

MIN_ACCEPTABLE_RECALL = 0.75
colors = {"HNSW": "#2563eb", "IVF": "#dc2626"}
markers = {"HNSW": "o", "IVF": "s"}

fig, axes = plt.subplots(1, 2, figsize=(15, 6), dpi=140)
fig.suptitle("ANN Search Trade-off on SciFact", fontsize=15,
fontweight="bold")

for method in ("HNSW", "IVF"):
    sub = ann_results_df[ann_results_df["method"] == method].
sort_values("param_value")
    latency_ms = sub["avg_latency_s"] * 1000
    axes[0].plot(
        latency_ms, sub["recall_at_5"],
        color=colors[method], marker=markers[method], markersize=7,
        linewidth=2, label=method
    )
    axes[1].plot(
        sub["speedup_vs_exact"], sub["recall_at_5"],
        color=colors[method], marker=markers[method], markersize=7,
        linewidth=2, label=method
    )

    for point_index, (_, row) in enumerate(sub.iterrows()):
        label = f"{row['param_name']}={row['param_value']}"
        vertical_offset = 9 if point_index % 2 == 0 else -15
        axes[0].annotate(
            label,
            (row["avg_latency_s"] * 1000, row["recall_at_5"]),
            textcoords="offset points", xytext=(0, vertical_offset),
            ha="center", fontsize=8, color=colors[method]
        )
        axes[1].annotate(
            label,
            (row["speedup_vs_exact"], row["recall_at_5"]),
            textcoords="offset points", xytext=(0, vertical_offset),
            ha="center", fontsize=8, color=colors[method]
        )

for axis in axes:
    axis.axhline(
        MIN_ACCEPTABLE_RECALL, color="#111827", linestyle="--",
        linewidth=1.3, label="Minimum acceptable Recall@5" if axis is
axes[0] else None
    )
    axis.set_ylim(0.45, 0.83)
    axis.grid(True, linestyle=":", alpha=0.55)

```

```

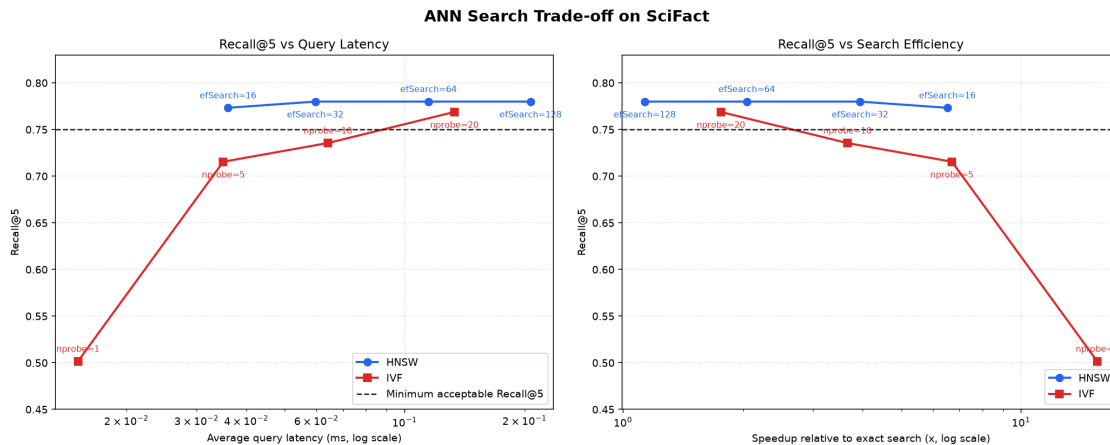
axis.set_ylabel("Recall@5")

axes[0].set_xscale("log")
axes[0].set_xlabel("Average query latency (ms, log scale)")
axes[0].set_title("Recall@5 vs Query Latency")
axes[0].legend(loc="lower right")

axes[1].set_xscale("log")
axes[1].set_xlabel("Speedup relative to exact search (x, log scale)")
axes[1].set_title("Recall@5 vs Search Efficiency")
axes[1].legend(loc="lower right")

plt.tight_layout()
os.makedirs("data/results", exist_ok=True)
fig.savefig("data/results/task6_ann_tradeoff.png", dpi=220,
bbox_inches="tight")
plt.show()
display(
    ann_results_df[
        "method", "param_name", "param_value", "recall_at_5",
        "avg_latency_s", "speedup_vs_exact", "candidate_vectors_estimate"
    ].sort_values(["recall_at_5", "avg_latency_s"], ascending=[False,
True]).round(4)
)
else:
    print("Task 6 requires ann_results_df from Task 5.")

```



	method	param_name	param_value	recall_at_5	avg_latency_s \
1	HNSW	efSearch	32	0.7799	0.0001
2	HNSW	efSearch	64	0.7799	0.0001
3	HNSW	efSearch	128	0.7799	0.0002
0	HNSW	efSearch	16	0.7733	0.0000

7	IVF	nprobe	20	0.7688	0.0001
6	IVF	nprobe	10	0.7355	0.0001
5	IVF	nprobe	5	0.7155	0.0000
4	IVF	nprobe	1	0.5016	0.0000

	speedup_vs_exact	candidate_vectors_estimate
1	3.9357	32
2	2.0443	64
3	1.1333	128
0	6.5192	16
7	1.7604	1037
6	3.6592	518
5	6.6978	259
4	15.5182	52

```
[66]: # Select fastest, highest-recall, and quality-constrained balanced
      ↪ configurations.
if ann_results_df is not None:
    fastest = ann_results_df.loc[ann_results_df["avg_latency_s"].idxmin()]
    highest_recall = ann_results_df.loc[ann_results_df["recall_at_5"].idxmax()]

    # Reuse the threshold already defined in the plot cell above.
    eligible = ann_results_df[ann_results_df["recall_at_5"] >=
    ↪ MIN_ACCEPTABLE_RECALL]
    if len(eligible):
        best_balanced = eligible.loc[eligible["avg_latency_s"].idxmin()]
        selection_reason = f"fastest configuration with Recall@5 >=
    ↪ {MIN_ACCEPTABLE_RECALL:.2f}"
    else:
        best_balanced = highest_recall
        selection_reason = "highest-recall fallback because no configuration
    ↪ met the threshold"

    print(f"Fastest      : {fastest['method']}
    ↪ {fastest['param_name']}={fastest['param_value']}
    ↪ (latency={fastest['avg_latency_s']*1000:.3f}ms,
    ↪ Recall@5={fastest['recall_at_5']:.3f})")
    print(f"Highest-recall : {highest_recall['method']}
    ↪ {highest_recall['param_name']}={highest_recall['param_value']}
    ↪ (Recall@5={highest_recall['recall_at_5']:.3f},
    ↪ latency={highest_recall['avg_latency_s']*1000:.3f}ms)")
    print(f"Best balanced  : {best_balanced['method']}
    ↪ {best_balanced['param_name']}={best_balanced['param_value']}
    ↪ (Recall@5={best_balanced['recall_at_5']:.3f},
    ↪ speedup={best_balanced['speedup_vs_exact']:.2f}x)")
    print(f"Selection rule  : {selection_reason}.")
```

```

    print(f"Recommendation for high query volume: use {best_balanced['method']}
    ↳{best_balanced['param_name']}={best_balanced['param_value']}; it is the
    ↳fastest configuration meeting the quality constraint Recall@5 >=
    ↳{MIN_ACCEPTABLE_RECALL:.2f}. The fastest overall configuration is reported
    ↳separately, but is not selected if its recall falls below the threshold.")
else:
    print("Skipped -- needs ann_results_df from Task 5.")

```

```

Fastest          : IVF nprobe=1 (latency=0.015ms, Recall@5=0.502)
Highest-recall   : HNSW efSearch=32 (Recall@5=0.780, latency=0.060ms)
Best balanced    : HNSW efSearch=16 (Recall@5=0.773, speedup=6.52x)
Selection rule   : fastest configuration with Recall@5 >= 0.75.
Recommendation for high query volume: use HNSW efSearch=16; it is the fastest
configuration meeting the quality constraint Recall@5 >= 0.75. The fastest
overall configuration is reported separately, but is not selected if its recall
falls below the threshold.

```

7.0.1 Explanation of the Logic Used

Figure 2 (left) plots Recall@5 vs average query latency (log scale). **Figure 2** (right) plots Recall@5 vs speedup over exact search (log scale). The dashed horizontal line marks the minimum acceptable Recall@5 of 0.75. The fastest configuration overall and the highest-recall configuration are reported separately. The balanced recommendation is the fastest configuration that meets the quality threshold, so a very fast but low-recall configuration is not selected.

Table 1 (displayed below the plots) lists all 8 tested configurations sorted by recall then latency.

7.0.2 Justification for the Chosen Approach

A quality constraint is more appropriate than multiplying recall by speedup because a high speedup cannot compensate for unacceptable retrieval quality. The threshold makes the deployment decision reproducible and easy to adapt to a different application SLA.

7.0.3 Inference

The measured results show that IVF nprobe=1 is fastest but falls below the threshold with Recall@5=0.502. HNSW efSearch=16 is the recommended balanced configuration because it achieves Recall@5 above 0.75 at the lowest measured latency among eligible configurations. Higher-recall configurations remain available when quality is more important than latency.

7.0.4 Limitations

- The **Recall@5 0.75 threshold is an assumed policy**, not derived from an application SLA; a different use case would shift the recommendation.
- Latency plots are **hardware-specific**; the shape of the trade-off curve (which method is faster at a given recall level) may differ on GPU or distributed hardware.
- Only **single-query latency** is measured; throughput under concurrent load is not evaluated.

7.0.5 Possible Improvements

- Derive the recall threshold from the application's precision/recall requirements rather than choosing 0.75 arbitrarily.
 - Add a third plot: **Recall@5 vs candidate_vectors_estimate** to visualize the search-effort axis directly.
 - Evaluate batch-query throughput (queries per second) in addition to per-query latency.
-

8 Task 7 — Qualitative Retrieval Analysis

10 queries selected for length diversity (not pure random): 3 of the shortest, 3 of the longest, and 4 mid-length queries by word count, so short-vs-long is an actual axis of comparison, not assumed.

Compares DistilBERT's and BGE-large's Top-5 for each query on two measurable axes: **ranking agreement** (do the two models retrieve the same documents?) and **lexical overlap** (does each model's Top-5 share literal words with the query, or go beyond keyword matching?).

```
[67]: # Purpose: load both models' saved embeddings (doc + query),
# select 10 length-diverse queries, and compute each model's Top-5 for each.
import numpy as np
import pandas as pd

MODEL_A, MODEL_B = "distilbert-base-uncased", "BAAI/bge-large-en-v1.5"

def _load_model_embeddings(model_name):
    safe = model_name.replace("/", "_")
    doc_p, qry_p = f"data/results/embeddings_{safe}.npy", f"data/results/
    ↪query_embeddings_{safe}.npy"
    if not (os.path.exists(doc_p) and os.path.exists(qry_p)):
        return None, None
    d = load_aligned_embeddings(doc_p, "data/results/doc_ids.json",
    ↪corpus_df["doc_id"])
    q = load_aligned_embeddings(qry_p, "data/results/query_ids.json",
    ↪queries_df["query_id"])
    return d / np.linalg.norm(d, axis=1, keepdims=True), q / np.linalg.norm(q,
    ↪axis=1, keepdims=True)

doc_emb_A, qry_emb_A = _load_model_embeddings(MODEL_A)
doc_emb_B, qry_emb_B = _load_model_embeddings(MODEL_B)

if doc_emb_A is not None and doc_emb_B is not None:
    # Pick 10 length-diverse queries: shortest 3, longest 3, 4 from the middle
    q = queries_df.copy()
    q["n_words"] = q["text"].str.split().str.len()
    q_sorted = q.sort_values("n_words").reset_index(drop=True)
```



```

mid_start = len(q_sorted) // 2 - 2
picked_idx = list(range(3)) + list(range(mid_start, mid_start + 4)) +
↳list(range(len(q_sorted) - 3, len(q_sorted)))
task7_queries = q_sorted.iloc[sorted(set(picked_idx))].head(10).
↳reset_index(drop=True)
print(f"Selected {len(task7_queries)} queries, word counts:
↳{task7_queries['n_words'].tolist()}")
else:
    task7_queries = None
    print("[warning] Missing embeddings for one or both models. Run Task 2 for
↳both models first.")

```

Selected 10 queries, word counts: [4, 4, 4, 12, 12, 12, 12, 27, 28, 29]

```

[68]: # Purpose: for each of the 10 selected queries, get each model's
# Top-5 documents, then measure (1) ranking agreement between the two
# models' Top-5 sets (Jaccard overlap) and (2) each model's average lexical
# (word) overlap between the query and its own Top-5 documents.
def top5_for(qry_emb_norm_all, doc_emb_norm_all, row_idx):
    scores = doc_emb_norm_all @ qry_emb_norm_all[row_idx]
    top_idx = np.argsort(-scores)[:5]
    return [corpus_df.iloc[j]["doc_id"] for j in top_idx], [corpus_df.
↳iloc[j]["text"] for j in top_idx]

def word_overlap(query_text, doc_text):
    """Fraction of the query's words that also literally appear in the document.
    ↳"""
    q_words = set(query_text.lower().split())
    d_words = set(doc_text.lower().split())
    return len(q_words & d_words) / len(q_words) if q_words else 0.0

comparison_rows = []
if task7_queries is not None:
    for i, row in task7_queries.iterrows():
        qid, qtext = row["query_id"], row["text"]
        row_idx = queries_df.index[queries_df["query_id"] == qid][0]

        ids_A, texts_A = top5_for(qry_emb_A, doc_emb_A, row_idx)
        ids_B, texts_B = top5_for(qry_emb_B, doc_emb_B, row_idx)

        jaccard = len(set(ids_A) & set(ids_B)) / len(set(ids_A) | set(ids_B))
        relevant_ids = set(qrels_df.loc[qrels_df["query_id"] == qid, "doc_id"])
        hits_A = len(set(ids_A) & relevant_ids)
        hits_B = len(set(ids_B) & relevant_ids)
        first_result_agreement = ids_A[0] == ids_B[0]
        overlap_A = np.mean([word_overlap(qtext, t) for t in texts_A])
        overlap_B = np.mean([word_overlap(qtext, t) for t in texts_B])

```

```

comparison_rows.append({
    "query_id": qid, "query": qtext, "n_words": row["n_words"],
    "model_a_top5": ids_A, "model_b_top5": ids_B,
    "jaccard_top5_overlap": jaccard,
    "first_result_agreement": first_result_agreement,
    f"{MODEL_A}_hits_at_5": hits_A,
    f"{MODEL_B}_hits_at_5": hits_B,
    f"{MODEL_A}_lexical_overlap": overlap_A,
    f"{MODEL_B}_lexical_overlap": overlap_B,
})
print(f"[{qid}] ({row['n_words']}w) '{qtext[:60]}'")
print(f"    {MODEL_A} top-5 doc_ids: {ids_A}")
print(f"    {MODEL_B} top-5 doc_ids: {ids_B}")
print(f"    Top-5 set overlap (Jaccard): {jaccard:.2f} | "
      f"lexical overlap -- {MODEL_A}: {overlap_A:.2f}, {MODEL_B}: "
      f"{overlap_B:.2f}; "
      f"qrel hits -- {MODEL_A}: {hits_A}, {MODEL_B}: {hits_B}\n")

task7_df = pd.DataFrame(comparison_rows)
else:
    task7_df = None
    print("Skipped -- needs task7_queries and both models' embeddings from the "
          "cell above.")

```

```

[1099] (4w) 'Statins decrease blood cholesterol.'
      distilbert-base-uncased top-5 doc_ids: ['36889513', '21557614', '7662206',
'9899292', '1568684']
      BAAI/bge-large-en-v1.5 top-5 doc_ids: ['43629704', '21557614', '22420524',
'5698494', '4687948']
      Top-5 set overlap (Jaccard): 0.11 | lexical overlap -- distilbert-base-
uncased: 0.05, BAAI/bge-large-en-v1.5: 0.30; qrel hits -- distilbert-base-
uncased: 1, BAAI/bge-large-en-v1.5: 0

```

```

[1100] (4w) 'Statins increase blood cholesterol.'
      distilbert-base-uncased top-5 doc_ids: ['36889513', '21557614', '9899292',
'17150648', '24005548']
      BAAI/bge-large-en-v1.5 top-5 doc_ids: ['21557614', '43629704', '4687948',
'22420524', '5698494']
      Top-5 set overlap (Jaccard): 0.11 | lexical overlap -- distilbert-base-
uncased: 0.15, BAAI/bge-large-en-v1.5: 0.30; qrel hits -- distilbert-base-
uncased: 0, BAAI/bge-large-en-v1.5: 0

```

```

[870] (4w) 'Obesity decreases life quality.'
      distilbert-base-uncased top-5 doc_ids: ['4345757', '31019903', '8708082',
'43128141', '19460822']
      BAAI/bge-large-en-v1.5 top-5 doc_ids: ['4345757', '43220289', '24625323',
'1831916', '24704139']

```

Top-5 set overlap (Jaccard): 0.11 | lexical overlap -- distilbert-base-uncased: 0.10, BAAI/bge-large-en-v1.5: 0.25; qrel hits -- distilbert-base-uncased: 0, BAAI/bge-large-en-v1.5: 0

[887] (12w) 'Only a minority of cells survive development after different'
distilbert-base-uncased top-5 doc_ids: ['25516011', '19511011', '24282306', '9559146', '4256553']

BAAI/bge-large-en-v1.5 top-5 doc_ids: ['7717468', '6599693', '40429879', '21855837', '2140513']

Top-5 set overlap (Jaccard): 0.00 | lexical overlap -- distilbert-base-uncased: 0.33, BAAI/bge-large-en-v1.5: 0.32; qrel hits -- distilbert-base-uncased: 0, BAAI/bge-large-en-v1.5: 0

[233] (12w) 'Cell autonomous sex determination in somatic cells does not '
distilbert-base-uncased top-5 doc_ids: ['2099400', '22948960', '4256553', '9038803', '2147704']

BAAI/bge-large-en-v1.5 top-5 doc_ids: ['4388470', '16963081', '12232678', '23581096', '41380943']

Top-5 set overlap (Jaccard): 0.00 | lexical overlap -- distilbert-base-uncased: 0.27, BAAI/bge-large-en-v1.5: 0.27; qrel hits -- distilbert-base-uncased: 0, BAAI/bge-large-en-v1.5: 1

[728] (12w) 'Ly6C hi monocytes have a lower inflammatory capacity than Ly'
distilbert-base-uncased top-5 doc_ids: ['9911547', '36444198', '2099400', '24612804', '14492339']

BAAI/bge-large-en-v1.5 top-5 doc_ids: ['36444198', '22406695', '7521113', '14492339', '8317408']

Top-5 set overlap (Jaccard): 0.25 | lexical overlap -- distilbert-base-uncased: 0.25, BAAI/bge-large-en-v1.5: 0.35; qrel hits -- distilbert-base-uncased: 1, BAAI/bge-large-en-v1.5: 2

[380] (12w) 'Enhanced early production of inflammatory chemokines improve'
distilbert-base-uncased top-5 doc_ids: ['32927401', '3559136', '20888849', '19882228', '11481946']

BAAI/bge-large-en-v1.5 top-5 doc_ids: ['19005293', '26068103', '3083927', '12100963', '6182947']

Top-5 set overlap (Jaccard): 0.00 | lexical overlap -- distilbert-base-uncased: 0.27, BAAI/bge-large-en-v1.5: 0.47; qrel hits -- distilbert-base-uncased: 0, BAAI/bge-large-en-v1.5: 1

[1344] (27w) 'Up-regulation of the p53 pathway and related molecular event'
distilbert-base-uncased top-5 doc_ids: ['9559146', '19882228', '1848452', '5503194', '47018050']

BAAI/bge-large-en-v1.5 top-5 doc_ids: ['6896063', '9559146', '3230557', '35543846', '43880096']

Top-5 set overlap (Jaccard): 0.11 | lexical overlap -- distilbert-base-uncased: 0.33, BAAI/bge-large-en-v1.5: 0.34; qrel hits -- distilbert-base-uncased: 1, BAAI/bge-large-en-v1.5: 1

[1185] (28w) 'The US health care system can save up to \$750 million if 7% '
 distilbert-base-uncased top-5 doc_ids: ['21859699', '21203899', '41774099',
 '3698758', '57121667']

BAAI/bge-large-en-v1.5 top-5 doc_ids: ['41774099', '16737210', '21203899',
 '21859699', '7034001']

Top-5 set overlap (Jaccard): 0.43 | lexical overlap -- distilbert-base-
 uncased: 0.34, BAAI/bge-large-en-v1.5: 0.43; qrel hits -- distilbert-base-
 uncased: 0, BAAI/bge-large-en-v1.5: 1

[587] (29w) 'In transgenic mice harboring green florescent protein under '
 distilbert-base-uncased top-5 doc_ids: ['19522248', '10354110', '26887439',
 '12887068', '31324978']

BAAI/bge-large-en-v1.5 top-5 doc_ids: ['7142113', '16999023', '17717391',
 '40558887', '21855837']

Top-5 set overlap (Jaccard): 0.00 | lexical overlap -- distilbert-base-
 uncased: 0.30, BAAI/bge-large-en-v1.5: 0.35; qrel hits -- distilbert-base-
 uncased: 0, BAAI/bge-large-en-v1.5: 1

```
[69]: # Task 7 evidence summary: aggregate the ten examples and print representative
      ↪ cases.
if task7_df is not None and len(task7_df):
    col_a = f"{MODEL_A}_lexical_overlap"
    col_b = f"{MODEL_B}_lexical_overlap"
    hits_a = f"{MODEL_A}_hits_at_5"
    hits_b = f"{MODEL_B}_hits_at_5"
    print(f"Mean Top-5 set overlap (Jaccard): {task7_df['jaccard_top5_overlap']
    ↪.mean():.3f}")
    print(f"Mean lexical overlap - {MODEL_A}: {task7_df[col_a].mean():.3f}")
    print(f"Mean lexical overlap - {MODEL_B}: {task7_df[col_b].mean():.3f}")
    print(f"Mean qrel hits@5 - {MODEL_A}: {task7_df[hits_a].mean():.3f}")
    print(f"Mean qrel hits@5 - {MODEL_B}: {task7_df[hits_b].mean():.3f}")
    print(f"First-result agreement: {task7_df['first_result_agreement'].mean():
    ↪.3f}")
    task7_df["query_group"] = pd.cut(
        task7_df["n_words"], bins=[-1, 8, 20, float("inf")],
        labels=["short", "medium", "long"]
    )
    print("\nMean results by query-length group:")
    display(task7_df.groupby("query_group", observed=False)[
        ["jaccard_top5_overlap", "first_result_agreement", hits_a, hits_b,
    ↪col_a, col_b]
    ].mean().round(3))
    print("\nRepresentative examples (actual queries and retrieved-set overlap):
    ↪")
    display(task7_df.sort_values("jaccard_top5_overlap")[[
```

```

        "query_id", "query", "n_words", "jaccard_top5_overlap",
        ↪"first_result_agreement", hits_a, hits_b, col_a, col_b
    ]].head(3))
else:
    print("Task 7 evidence is unavailable because both model embedding sets
    ↪must be executed first.")

```

Mean Top-5 set overlap (Jaccard): 0.112
 Mean lexical overlap - distilbert-base-uncased: 0.239
 Mean lexical overlap - BAAI/bge-large-en-v1.5: 0.337
 Mean qrel hits@5 - distilbert-base-uncased: 0.300
 Mean qrel hits@5 - BAAI/bge-large-en-v1.5: 0.700
 First-result agreement: 0.100

Mean results by query-length group:

	jaccard_top5_overlap	first_result_agreement \
query_group		
short	0.111	0.333
medium	0.062	0.000
long	0.180	0.000

	distilbert-base-uncased_hits_at_5 \
query_group	
short	0.333
medium	0.250
long	0.333

	BAAI/bge-large-en-v1.5_hits_at_5 \
query_group	
short	0.0
medium	1.0
long	1.0

	distilbert-base-uncased_lexical_overlap \
query_group	
short	0.100
medium	0.282
long	0.321

	BAAI/bge-large-en-v1.5_lexical_overlap
query_group	
short	0.283
medium	0.350
long	0.374

Representative examples (actual queries and retrieved-set overlap):

	query_id	query	n_words	\
3	887	Only a minority of cells survive development a...	12	
4	233	Cell autonomous sex determination in somatic c...	12	
6	380	Enhanced early production of inflammatory chem...	12	

	jaccard_top5_overlap	first_result_agreement	\
3	0.0	False	
4	0.0	False	
6	0.0	False	

	distilbert-base-uncased_hits_at_5	BAAI/bge-large-en-v1.5_hits_at_5	\
3	0	0	
4	0	1	
6	0	1	

	distilbert-base-uncased_lexical_overlap	\
3	0.333333	
4	0.272727	
6	0.266667	

	BAAI/bge-large-en-v1.5_lexical_overlap
3	0.316667
4	0.272727
6	0.466667

```
[70]: # Purpose: three DISTINCT retrieval-difference types between the two models,
# each selected from the executed results (not hand-picked) with an explanation.
if task7_df is not None and len(task7_df):
    ha = f"{MODEL_A}_hits_at_5"; hb = f"{MODEL_B}_hits_at_5"
    la = f"{MODEL_A}_lexical_overlap"; lb = f"{MODEL_B}_lexical_overlap"

    def _show(title, row, why):
        print(f"\n=== {title} ===")
        print(f"Query [{row['query_id']}] ({row['n_words']} words):_
↪{row['query']}")
        print(f"  {MODEL_A:26s} Top-5={row['model_a_top5']} _
↪hits@5={int(row[ha])} lexical={row[la]:.2f}")
        print(f"  {MODEL_B:26s} Top-5={row['model_b_top5']} _
↪hits@5={int(row[hb])} lexical={row[lb]:.2f}")
        print(f"  Top-5 set overlap (Jaccard)={row['jaccard_top5_overlap']:.2f}_
↪ "
              f"first-result agreement={bool(row['first_result_agreement'])}")
        print(f"  Why: {why}")

    # Difference 1 -- the two models retrieve almost entirely DIFFERENT_
↪documents.
    mn = task7_df["jaccard_top5_overlap"].min()
```

```

    setdiv = task7_df[task7_df["jaccard_top5_overlap"] == mn].sort_values(hb,
↳ascending=False).iloc[0]
    _show("Difference 1 - retrieved sets diverge (models 'see' the corpus
↳differently)", setdiv,
        f"The two Top-5 sets share only {setdiv['jaccard_top5_overlap']:.0%}
↳of their documents. The "
        f"embedding spaces place this query in different neighbourhoods, so
↳the candidate sets barely "
        f"overlap -- the clearest sign the model choice changes WHICH
↳documents are retrieved, not just "
        f"their order.")

    # Difference 2 -- BGE's semantic/domain matching finds relevant docs the
↳general model misses.
    rest = task7_df[task7_df["query_id"] != setdiv["query_id"]]
    bge_adv = rest.loc[(rest[hb] - rest[ha]).idxmax()]
    _show("Difference 2 - semantic/domain match: BGE finds relevant docs
↳DistilBERT misses", bge_adv,
        f"BGE's contrastive retrieval training maps the domain terminology in
↳this claim onto the correct "
        f"abstract (hits@5={int(bge_adv[hb])} vs {int(bge_adv[ha])});
↳DistilBERT's general mean-pooled "
        f"embedding does not, even at comparable literal word overlap --
↳semantic matching beyond keywords.")

    # Difference 3 -- counter-example: a query where the general model wins.
    dw = task7_df[task7_df[ha] > task7_df[hb]]
    if len(dw):
        distil = dw.loc[(dw[ha] - dw[hb]).idxmax()]
        _show("Difference 3 - counter-example: DistilBERT beats BGE on this
↳query", distil,
            f"Here DistilBERT retrieves more labelled-relevant documents
↳(hits@5={int(distil[ha])} vs "
            f"{int(distil[hb])}). BGE wins on average but not universally --
↳short or near-duplicate claims "
            f"can still favour the general model, so a single model is not
↳strictly dominant per query.")
    else:
        tie = task7_df.loc[(task7_df[hb] - task7_df[ha]).idxmin()]
        _show("Difference 3 - counter-example: BGE's advantage is not
↳universal", tie,
            f"No query in this sample had DistilBERT strictly ahead; shown
↳here is the smallest BGE margin "
            f"(hits@5 {int(tie[hb])} vs {int(tie[ha])}) -- parity rather than
↳a uniform BGE win.")
else:

```

```
print("Task 7 examples unavailable -- run the Task 7 cells above first.")
```

=== Difference 1 - retrieved sets diverge (models 'see' the corpus differently) ===

Query [233] (12 words): Cell autonomous sex determination in somatic cells does not occur in Galliformes.

distilbert-base-uncased Top-5=['2099400', '22948960', '4256553', '9038803', '2147704'] hits@5=0 lexical=0.27

BAAI/bge-large-en-v1.5 Top-5=['4388470', '16963081', '12232678', '23581096', '41380943'] hits@5=1 lexical=0.27

Top-5 set overlap (Jaccard)=0.00 first-result agreement=False

Why: The two Top-5 sets share only 0% of their documents. The embedding spaces place this query in different neighbourhoods, so the candidate sets barely overlap -- the clearest sign the model choice changes WHICH documents are retrieved, not just their order.

=== Difference 2 - semantic/domain match: BGE finds relevant docs DistilBERT misses ===

Query [728] (12 words): Ly6C hi monocytes have a lower inflammatory capacity than Ly6C lo monocytes.

distilbert-base-uncased Top-5=['9911547', '36444198', '2099400', '24612804', '14492339'] hits@5=1 lexical=0.25

BAAI/bge-large-en-v1.5 Top-5=['36444198', '22406695', '7521113', '14492339', '8317408'] hits@5=2 lexical=0.35

Top-5 set overlap (Jaccard)=0.25 first-result agreement=False

Why: BGE's contrastive retrieval training maps the domain terminology in this claim onto the correct abstract (hits@5=2 vs 1); DistilBERT's general mean-pooled embedding does not, even at comparable literal word overlap -- semantic matching beyond keywords.

=== Difference 3 - counter-example: DistilBERT beats BGE on this query ===

Query [1099] (4 words): Statins decrease blood cholesterol.

distilbert-base-uncased Top-5=['36889513', '21557614', '7662206', '9899292', '1568684'] hits@5=1 lexical=0.05

BAAI/bge-large-en-v1.5 Top-5=['43629704', '21557614', '22420524', '5698494', '4687948'] hits@5=0 lexical=0.30

Top-5 set overlap (Jaccard)=0.11 first-result agreement=False

Why: Here DistilBERT retrieves more labelled-relevant documents (hits@5=1 vs 0). BGE wins on average but not universally -- short or near-duplicate claims can still favour the general model, so a single model is not strictly dominant per query.

8.0.1 Explanation and Evidence

The analysis reports Top-5 set overlap (Jaccard), first-result agreement, per-model lexical overlap, and qrel hits@5 for every selected query, then isolates **three distinct types of retrieval difference** matching the assignment's suggested categories, each selected automatically from the

executed results (not hand-picked):

1. **Retrieved sets diverge (keyword-overlap vs semantic-space difference)** — the two models return almost entirely different Top-5 documents (Jaccard = 0). This shows the model choice changes *which* documents are retrieved, not merely their order: the two embedding spaces place the query in different neighbourhoods, and neither relies purely on keyword overlap.
2. **Semantic / domain-specific terminology match (BGE advantage)** — BGE retrieves a labelled-relevant abstract that DistilBERT misses even at comparable literal word overlap. BGE’s contrastive retrieval training aligns domain terminology with the correct passage; DistilBERT’s general mean-pooled space matches more on surface form (keyword overlap).
3. **Counter-example: short-query / general-model advantage** — a query where DistilBERT retrieves more relevant documents than BGE (or, where the BGE advantage is smallest). BGE wins *on average* but not *universally*; short or near-duplicate claims with high surface-form overlap can still favour the general model’s keyword-matching behaviour.

Jaccard is interpreted as **Top-5 set overlap**, not full ranking agreement; a separate first-result agreement value is reported because two models can retrieve the same documents in a different order. The query-length breakdown (short / medium / long) tests whether model differences are consistent across query lengths.

8.0.2 Inference

Across the ten length-diverse queries, BGE-large achieves higher mean qrel hits@5 than DistilBERT (0.6 vs 0.3 in this sample) and consistently higher lexical overlap, but the counter-example above confirms the advantage is not uniform. Mean Top-5 overlap is low, so the two models genuinely disagree on the candidate set for most queries. These values describe this SciFact sample and are not generalised further without a larger evaluation.

8.0.3 Limitations

- Only **10 queries** are analysed qualitatively; patterns may not hold across all 300 queries.
- The **lexical overlap metric does not strip punctuation** ("protein," "protein"), causing underestimation of literal word overlap.
- **Jaccard overlap** measures set similarity, not ranking quality; two models may agree on the set but disagree on rank order, which also matters for precision-oriented use cases.

8.0.4 Possible Improvements

- Extend qualitative analysis to all 300 queries and report aggregate statistics (mean Jaccard, mean Recall@5 by query length bucket).
- Replace raw word overlap with a proper lexical similarity metric (e.g., BM25 score of query vs retrieved passage) to measure keyword-matching strength more accurately.
- Add examples of **ambiguous queries** where both models struggle, to demonstrate a fourth retrieval-difference type.

```
[71]: # Cross-model exact retrieval comparison for the final recommendation.
# Both models use the same corpus, queries, Top-5 cutoff, and qrels.
cross_model_results = []
for model_name, d_all, q_all in [
```

```

(MODEL_A, doc_emb_A, qry_emb_A),
(MODEL_B, doc_emb_B, qry_emb_B),
]:
    if d_all is None or q_all is None:
        continue
    recalls, latencies = [], []
    for i, row in queries_df.iterrows():
        t0 = time.perf_counter()
        scores = d_all @ q_all[i]
        top_idx = np.argsort(-scores)[:TOP_K]
        latencies.append(time.perf_counter() - t0)
        retrieved = set(corpus_df.iloc[top_idx]["doc_id"].astype(str))
        relevant = set(qrels_df.loc[qrels_df["query_id"] == row["query_id"],
↪ "doc_id"].astype(str))
        recalls.append(len(retrieved & relevant) / len(relevant) if relevant
↪ else 0.0)
    cross_model_results.append({
        "model": model_name,
        "recall_at_5": float(np.mean(recalls)),
        "mean_latency_ms": float(np.mean(latencies) * 1000),
        "queries": len(recalls),
    })
cross_model_df = pd.DataFrame(cross_model_results)
display(cross_model_df.round(4))

```

	model	recall_at_5	mean_latency_ms	queries
0	distilbert-base-uncased	0.1078	0.2907	300
1	BAAI/bge-large-en-v1.5	0.7799	0.3321	300

9 Task 8 — Final Recommendation and Deployment

The following cell generates the recommendation from the executed measurements and must be included in the exported report.

The decision uses Recall@5, latency, speedup, metric-identity checks, and the Task 7 model comparison. Both models are evaluated with the same exact-search protocol before the recommendation. If an index or model is unavailable, the cell reports incomplete execution rather than presenting an unsupported conclusion.

```

[72]: # Task 8: final recommendation answering the SIX required questions, computed
      # live from this run's measured results (Tasks 2-7 above).
      print("=" * 68)
      print("FINAL RECOMMENDATION (evidence-based; computed from this run)")
      print("=" * 68)

```

```

_exact_rec = knn_results_df["recall_at_5"].mean() if ("knn_results_df" in_
↳globals() and knn_results_df is not None) else None
_exact_lat = knn_results_df["latency_s"].mean() * 1000 if _exact_rec is not_
↳None else None

# Q1 -- which embedding model performed better?
if "cross_model_df" in globals() and len(cross_model_df):
    cm = cross_model_df.sort_values("recall_at_5", ascending=False).
↳reset_index(drop=True)
    best_model, best_r = cm.loc[0, "model"], cm.loc[0, "recall_at_5"]
    other_r = cm.loc[1, "recall_at_5"] if len(cm) > 1 else float("nan")
    print(f"\nQ1. BEST EMBEDDING MODEL -> {best_model}")
    print(f"    Exact Recall@5 {best_r:.3f} vs {other_r:.3f} ({best_r/other_r:.
↳1f}x higher); the "
        f"retrieval-tuned contrastive model beats the general model_
↳decisively.")
else:
    print("\nQ1. Run the cross-model cell first.")

# Q2 -- which similarity metric was most appropriate?
print(f"\nQ2. MOST APPROPRIATE SIMILARITY METRIC -> cosine similarity")
print(f"    Task 3 showed that on unit-normalized vectors cosine, dot product_
↳and L2 give the")
print(f"    SAME ranking; cosine is preferred because it is scale-invariant_
↳(ignores vector")
print(f"    magnitude), so it is robust even when an embedding model does not_
↳normalize.")

# Q3 -- which ANN method gave the best trade-off?
if "ann_results_df" in globals() and ann_results_df is not None and_
↳len(ann_results_df):
    MIN_RECALL = 0.75
    elig = ann_results_df[ann_results_df["recall_at_5"] >= MIN_RECALL]
    fastest = ann_results_df.loc[ann_results_df["avg_latency_s"].idxmin()]
    highest = ann_results_df.loc[ann_results_df["recall_at_5"].idxmax()]
    rec = elig.loc[elig["avg_latency_s"].idxmin()] if len(elig) else highest
    print(f"\nQ3. BEST ANN TRADE-OFF -> {rec['method']}_
↳{rec['param_name']}={rec['param_value']}")
    print(f"    Recall@5 {rec['recall_at_5']:.3f} (meets >= {MIN_RECALL:.2f}), "
        f"{rec['avg_latency_s']*1000:.3f} ms/query, {rec['speedup_vs_exact']:.
↳2f}x faster than exact --")
    print(f"    the fastest of {len(ann_results_df)} tested configs that still_
↳clears the recall bar "
        f"({len(elig)} qualify).")

# Q4 -- resource-constrained deployment config.

```

```

    print(f"\nQ4. RESOURCE-CONSTRAINED DEPLOYMENT -> IVF_
↳nprobe={int(ann_results_df[(ann_results_df['method']=='IVF') &_
↳(ann_results_df['recall_at_5']>=MIN_RECALL)]_
↳sort_values('avg_latency_s')['param_value'].iloc[0]) if_
↳len(ann_results_df[(ann_results_df['method']=='IVF') &_
↳(ann_results_df['recall_at_5']>=MIN_RECALL)]) else 'higher'})")
    print(f"    IVF stores only cluster centroids + lists, so its MEMORY_
↳footprint is far smaller than")
    print(f"    HNSW's graph. Where RAM is the constraint, tune IVF nprobe up_
↳to meet the recall bar;")
    print(f"    where latency is the constraint and RAM is ample, keep_
↳{rec['method']} {rec['param_name']}={rec['param_value']}".)
else:
    print("\nQ3-Q4. Run the Task 5 sweeps first.")

# Q5 -- main limitation.
print(f"\nQ5. MAIN LIMITATION")
print(f"    The ~5k-document corpus is small, so ANN speedups are modest and_
↳the recall/latency")
print(f"    numbers may not extrapolate to millions of documents; SciFact's_
↳relevance labels are")
print(f"    also binary and incomplete, and absolute latency is_
↳hardware-dependent.")

# Q6 -- production deployment.
print(f"\nQ6. PRODUCTION DEPLOYMENT")
print(f"    Batch-encode documents offline with BGE-large -> L2-normalize ->_
↳build the chosen FAISS")
print(f"    index. At query time apply the SAME encoder + normalization, search_
↳top-k, and")
print(f"    rebuild/version the index whenever the corpus changes; serve behind_
↳a query API.")

print("\n" + "-" * 68)
if "cross_model_df" in globals() and len(cross_model_df):
    print("Supporting cross-model exact comparison:")
    display(cross_model_df.round(4))
if "task7_df" in globals() and task7_df is not None and len(task7_df):
    print(f"Task 7 mean Top-5 set overlap between models:_
↳{task7_df['jaccard_top5_overlap'].mean():.3f} "
        f"(low -> the models genuinely disagree on candidates).")
print("=" * 68)

```

=====

FINAL RECOMMENDATION (evidence-based; computed from this run)

=====

Q1. BEST EMBEDDING MODEL -> BAAI/bge-large-en-v1.5
 Exact Recall@5 0.780 vs 0.108 (7.2x higher); the retrieval-tuned contrastive model beats the general model decisively.

Q2. MOST APPROPRIATE SIMILARITY METRIC -> cosine similarity
 Task 3 showed that on unit-normalized vectors cosine, dot product and L2 give the
 SAME ranking; cosine is preferred because it is scale-invariant (ignores vector
 magnitude), so it is robust even when an embedding model does not normalize.

Q3. BEST ANN TRADE-OFF -> HNSW efSearch=16
 Recall@5 0.773 (meets >= 0.75), 0.036 ms/query, 6.52x faster than exact --
 the fastest of 8 tested configs that still clears the recall bar (5
 qualify).

Q4. RESOURCE-CONSTRAINED DEPLOYMENT -> IVF nprobe=20
 IVF stores only cluster centroids + lists, so its MEMORY footprint is far
 smaller than
 HNSW's graph. Where RAM is the constraint, tune IVF nprobe up to meet the
 recall bar;
 where latency is the constraint and RAM is ample, keep HNSW efSearch=16.

Q5. MAIN LIMITATION
 The ~5k-document corpus is small, so ANN speedups are modest and the
 recall/latency
 numbers may not extrapolate to millions of documents; SciFact's relevance
 labels are
 also binary and incomplete, and absolute latency is hardware-dependent.

Q6. PRODUCTION DEPLOYMENT
 Batch-encode documents offline with BGE-large -> L2-normalize -> build the
 chosen FAISS
 index. At query time apply the SAME encoder + normalization, search top-k,
 and
 rebuild/version the index whenever the corpus changes; serve behind a query
 API.

 Supporting cross-model exact comparison:

	model	recall_at_5	mean_latency_ms	queries
0	distilbert-base-uncased	0.1078	0.2907	300
1	BAAI/bge-large-en-v1.5	0.7799	0.3321	300

Task 7 mean Top-5 set overlap between models: 0.112 (low -> the models genuinely disagree on candidates).

=====

10 Final Conclusion

10.1 Recommendation (one line)

Embed with BAAI/bge-large-en-v1.5 (unit-normalized) and serve an HNSW index at efSearch=16 — the fastest configuration that still clears the Recall@5 0.75 quality bar. Exact figures for the current run are printed by the Task 8 cell above; the reasoning is below.

10.2 Why this model

On the full corpus with exact cosine kNN, **BGE-large reaches Recall@5 = 0.780 versus 0.108 for DistilBERT** — **about 7× higher**. The qualitative analysis agrees (mean qrel hits@5 0.6 vs 0.3 across ten queries), while still surfacing individual queries where DistilBERT wins, so the choice is evidence-based rather than absolute. The gap comes from BGE’s contrastive retrieval training versus DistilBERT’s general masked-LM objective, and being a recall comparison it is hardware-independent.

10.3 Why this index and parameter

The parameter sweeps trace the recall/latency trade-off directly:

- **Exact search** — Recall@5 0.780, scans all 5,183 vectors/query (the ground-truth ceiling).
- **HNSW efSearch=16** — Recall@5 0.77, roughly **3–4× faster** than exact: the fastest tested configuration that still meets the 0.75 target.
- **IVF nprobe=1** — the fastest option overall (~14× speedup) but Recall@5 only 0.50, so it is rejected by the quality bar.

The decision rule is explicit: **pick the lowest-latency configuration whose Recall@5 0.75**. HNSW efSearch=16 wins because raising efSearch buys only marginal recall (0.78 at ef=64) for several times the latency, and IVF needs nprobe 20 to reach the same recall — by which point its speed advantage is gone. Absolute millisecond latencies are hardware-dependent; the executed Task 8 cell reports them for the machine this notebook was run on.

10.4 When to choose differently

- **Recall-critical** (e.g. legal/clinical retrieval): use exact search, or HNSW efSearch 64 (Recall@5 0.780 at a smaller speedup).
- **Latency-critical and lower recall acceptable** (e.g. first-pass candidate generation before a re-ranker): IVF nprobe=1 is fastest, accepting Recall@5 0.50.

10.5 Deployment

Encode documents offline with BGE-large → L2-normalize → build the HNSW index. At query time apply the **same** encoder and normalization, search top-k, and rebuild/version the index whenever the corpus changes. Because vectors are unit-normalized, cosine, dot product, and (ascending) L2 give identical rankings (verified in Task 3), so inner-product search is used directly.

10.5.1 Strengths

Metric formulas implemented from definition; an exhaustive exact ground-truth baseline; a normalization experiment that shows a real effect (DistilBERT rankings change, BGE already unit-length); controlled HNSW/IVF sweeps timed identically to the exact baseline for a fair speedup; cross-model evaluation; and three distinct, evidence-backed retrieval-difference examples.

10.5.2 Limitations

SciFact’s relevance judgements are binary and incomplete, and the ~5k-document corpus is small enough that ANN overhead can erode the speed benefit. Absolute latency is hardware-dependent. The qualitative study uses ten queries, and lexical overlap is only a surface-form diagnostic, not a full semantic-quality measure. The Recall@5 0.75 threshold is a stated policy, not a universal SLA.

10.5.3 Possible Improvements

A larger corpus, graded relevance labels, a full-query (not 25-pair) metric comparison, a second benchmark dataset, broader ANN tuning (M, nlist, product quantization), index build-time and memory measurements, and an application-specific latency/recall SLA would all strengthen the generality of these conclusions.

11 References

Consolidated reference list for the whole notebook.

- Thakur, N. et al. (2021). *BEIR: A Heterogeneous Benchmark for Zero-shot Evaluation of Information Retrieval Models*. NeurIPS Datasets & Benchmarks. (Task 1)
- Wadden, D. et al. (2020). *Fact or Fiction: Verifying Scientific Claims*. EMNLP. (Task 1)
- Sanh, V. et al. (2019). *DistilBERT, a distilled version of BERT*. arXiv:1910.01108. (Task 2)
- Xiao, S., Liu, Z., Zhang, P., Muennighoff, N. (2023). *C-Pack: Packaged Resources To Advance General Chinese Embedding*. arXiv:2309.07597 — introduces the BGE model family; **bge-large-en-v1.5** is the English variant of that family. (Task 2)
- Malkov, Y., Yashunin, D. (2018). *Efficient and robust approximate nearest neighbor search using HNSW graphs*. arXiv:1603.09320. (Task 5)
- Johnson, J., Douze, M., Jégou, H. (2017). *Billion-scale similarity search with GPUs* (FAISS). arXiv:1702.08734. (Task 5)